

EBM TRANSFORMER COMPOSITIONAL GRASPING: A FROZEN MUJoCo STRESS TEST AND NEGATIVE RE- SULT

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ABSTRACT

Compositional grasping is an attractive idea: instead of learning one opaque scalar score for every grasp, a robot should compose object geometry, contact stability, task compatibility, clutter risk, feasibility, and learned context. This paper tests that idea under a deliberately hostile protocol. We rebuild an energy-based Transformer grasp ranker as a CPU-only MuJoCo benchmark with rollout-labeled training data, analytic force-closure baselines, CEM-style grasp search, learned MLP/HGB/Transformer/ensemble rankers, a calibrated stacked ranker, a robust perturbation baseline, hostile ablations, stress sweeps, seed-level paired statistics, and pre-specified negative cases. The final frozen evidence contains 1152 training rollouts, 18480 main evaluation rows, 4680 ablation rows, and 4320 stress rows. The terminal decision is **KILL ARCHIVE**. The reason is not formatting, lack of effort, or a missing plot: v5 does not beat strongest non-oracle aggregate baseline ensemble_uncertainty_ranker by 0.030; v5 does not beat strongest combined-shift baseline calibrated_stacked_ranker by 0.030; ablation gate fails because monolithic_scalar_energy_only, no_cem_fallback, no_collision_energy matches or beats full v5; maximum stress gate fails against ensemble_uncertainty_ranker. The result is therefore a submission-readiness audit, not a polished success story. It shows exactly where the compositional EBM/Transformer mechanism fails to separate itself from strong analytic and learned alternatives.

1 WHY THIS PAPER EXISTS

Robotic grasping is full of mechanisms that sound correct before they are placed under strong baselines. Analytic grasp metrics encode contact geometry and wrench resistance. Large synthetic grasp datasets and grasp-quality networks showed that analytic labels can train useful grasp planners at scale (Mahler et al., 2017). Large benchmarks such as GraspNet-1Billion pushed the field toward dense 6-DoF evaluation in clutter (Fang et al., 2020). Contact-based 6-DoF proposal networks further showed that direct depth-to-grasp distributions can be practical in structured clutter (Sundermeyer et al., 2021). At the same time, vision-based closed-loop grasping and manipulation systems show that learned policies can dominate simple hand-designed selectors when data and feedback are sufficient (Kalashnikov et al., 2018; Zeng et al., 2018).

The hypothesis behind this project is narrower than “Transformers solve grasping.” It says that grasp selection under object, contact, task, and clutter shift should benefit from an explicit additive energy decomposition. The claim is plausible. Energy-based models provide a natural language for ranking structured outputs (LeCun et al., 2006; Grathwohl et al., 2020). Transformers provide a compact architecture for contextual sequence scoring (Vaswani et al., 2017). Task-conditioned manipulation work also suggests that semantic and spatial factors should not be collapsed too early (Detry et al., 2017; Zeng et al., 2021; Shridhar et al., 2022).

Plausible is not enough. The useful question is whether explicit compositional energy survives a strong closed-loop grasp-selection audit. The answer in this frozen rebuild is no. The method improves the old ranker in some places, but it does not clear the strongest non-oracle baselines, and its own ablations expose that the mechanism is not identified cleanly.

2 PROBLEM SETTING

Each episode samples a source-to-target grasping scene s and a candidate set $A(s) = \{a_1, \dots, a_m\}$. A candidate action a is a planar parallel-jaw grasp specified by center, angle, jaw width, closing force, and lift tilt. The MuJoCo rollout returns success $Y(s, a)$ and diagnostics: drop, slip, collision, unsafe force, energy, task satisfaction, and final displacement. The learning system observes analytic and learned features, selects one candidate, and then the simulator executes the grasp.

The paper evaluates a ranker, not a full perception stack. The benchmark abstracts perception into object extents, shape family, friction proxy, task labels, clutter geometry, and candidate features. This is intentional. If a compositional grasp score cannot beat strong rankers in this controlled setting, it is not ready for a larger visual or hardware claim. If it does beat them, the next version would still need RGB-D, tactile/material uncertainty, hardware, and public-benchmark validation.

2.1 SHIFTS

The frozen main protocol contains eleven splits: seen simple objects, unseen dimensions, unseen shape family, slippery contact, clutter collision, task constraint shift, combined composition shift, long high-aspect bars, thin handle shift, adversarial clutter gap, and material-shift proxy. The last five are the hostile splits. They deliberately combine factors that make naive compositional scoring brittle: narrow viable approach windows, collision-prone clutter, low friction, high aspect ratio, and task-specific contact constraints.

2.2 METHODS

The comparison set is intentionally uncomfortable for the proposed method. It includes: `random_grasp`; `antipodal_geometry`; `force_closure_score`; `collision_aware_force_closure`; `cem_grasp_search`; `robust_perturbation_ranker`; `mlp_energy_model`; `gradient_boosted_energy_model`; `transformer_policy_ranker`; `ensemble_uncertainty_ranker`; `calibrated_stacked_ranker`; the old compositional EBM; the repaired EBM-v5; and a non-submission oracle grid. The oracle evaluates multiple candidate grasps in MuJoCo and is included only to show whether the candidate set contains a better option.

3 THE COMPOSITIONAL RANKER

Let $c_k(s, a)$ denote analytic costs for object alignment, contact stability, task compatibility, collision clearance, feasibility, and robust perturbation risk. Let $p_T(s, a)$ be the Transformer score, $p_M(s, a)$ the MLP score, $p_H(s, a)$ the gradient-boosted score, $p_E(s, a)$ the ensemble score, and $p_S(s, a)$ the scalar-energy score. The old method used a simple learned-context minus weighted-energy objective. The repaired v5 method uses

$$R_{v5}(s, a) = 1.35 p_{\text{stack}}(s, a) - \sum_k \lambda_k c_k(s, a) - h(s, a), \quad (1)$$

where

$$p_{\text{stack}} = 0.30 p_T + 0.26 p_M + 0.24 p_H + 0.14 p_E + 0.06 p_S. \quad (2)$$

The hard penalty h activates when collision or robust perturbation risk crosses a fixed threshold. If the selected candidate has high robust risk, v5 falls back to the robust perturbation analytic ranker. This fallback is not hidden; it is an explicit part of the method and is ablated.

The fallback is also a weakness. If the fallback dominates the decision, then the paper is not proving that compositional EBM scoring is necessary. It is proving that a robust analytic fallback is useful. That distinction becomes central in the ablation results.

4 WHAT THE THEORY CAN AND CANNOT SAY

The theory here is intentionally narrow. It exists to connect the method to falsifiable assumptions, not to turn a simulator result into a hardware theorem.

Proposition 1: additive certificates. Assume each normalized energy $c_k(s, a)$ upper-bounds a corresponding failure probability contribution up to calibration error ϵ_k :

$$\Pr[F_k \mid s, a] \leq c_k(s, a) + \epsilon_k.$$

Then by the union bound,

$$\Pr[\cup_k F_k \mid s, a] \leq \sum_k c_k(s, a) + \sum_k \epsilon_k.$$

Thus, if the energies are calibrated and non-redundant, minimizing a positive weighted sum can select a candidate with lower certified failure risk.

Proposition 2: learned context cannot rescue a bad candidate set. If no candidate in $A(s)$ satisfies the required contact, clearance, and task constraints, every ranker over $A(s)$ fails. The oracle grid in this paper is therefore not a submission baseline; it is a diagnostic that asks whether the candidate set itself has a viable grasp. When the oracle remains weak, better energy composition is not the bottleneck.

Proposition 3: ablation necessity. If removing a claimed component c_j does not reduce success or worsen safety under the hostile splits, then the frozen evidence does not support c_j as a necessary mechanism. This does not prove c_j is useless in all worlds. It proves only that this paper cannot claim it as the reason for success.

Negative theorem: redundancy under fallback. Suppose the final decision rule is

$$a^* = \begin{cases} \arg \max_a R_{v5}(s, a), & \rho(s, a) \leq \tau, \\ \arg \max_a R_{\text{robust}}(s, a), & \rho(s, a) > \tau. \end{cases}$$

If the hostile split places most selected candidates in the second branch, then component ablations inside R_{v5} can become observationally irrelevant. In that regime, the paper is testing robust fallback more than compositional energy. This is exactly why the ablation gate is severe.

5 FROZEN PROTOCOL

The frozen command is recorded in `docs/paper68_protocol_freeze_20260620.md`. No method tuning is permitted after that document. The development log records two pre-freeze checks. The first used a hard robust fallback and produced the strongest v5 behavior. The second replaced the fallback with a normalized blend, which worsened combined-shift success and safety. The frozen method therefore uses the hard fallback. This is not result polishing after the freeze; it is the final pre-freeze design choice.

The frozen gates are:

1. v5 must beat the strongest non-oracle aggregate baseline by at least 0.030 success.
2. v5 must beat the strongest non-oracle combined-composition baseline by at least 0.030 success.
3. v5 safety failures must not exceed the strongest aggregate non-oracle baseline by more than 0.020.
4. No removed-component ablation may match or beat full v5 within 0.020 success.
5. v5 must not lose the maximum-stress gate to a non-oracle baseline.

Passing all local gates would still imply only STRONG_REVISION because this is not hardware and not a recognized public grasp benchmark. Failing any gate implies KILL_ARCHIVE.

6 MAIN RESULTS

Table 1 is the main aggregate gate. The value of this table is not that it makes the proposed method look good. Its value is that it places v5 beside analytic, robust, learned, stacked, old-method, and oracle alternatives at the same evidence scale. The decision macro records the result: KILL_ARCHIVE.

Table 1: Aggregate frozen results across all grasping shifts. Success is higher-is-better; drop, collision, unsafe force, and energy are lower-is-better.

Method	Episodes	Success	Slip	Drop	Collision	Unsafe	Energy
Random	1320	0.274	0.078	0.535	0.262	0.000	1.784
Antipodal	1320	0.295	0.074	0.388	0.226	0.000	2.603
Force closure	1320	0.438	0.067	0.389	0.229	0.000	2.410
Collision-aware FC	1320	0.426	0.068	0.411	0.225	0.000	2.479
CEM analytic	1320	0.447	0.068	0.368	0.225	0.000	2.498
Robust perturb.	1320	0.410	0.070	0.436	0.227	0.000	2.444
MLP energy	1320	0.689	0.053	0.189	0.223	0.000	2.171
HGB energy	1320	0.747	0.054	0.145	0.217	0.000	2.268
Transformer	1320	0.679	0.055	0.176	0.227	0.000	2.223
Ensemble	1320	0.747	0.058	0.141	0.217	0.000	2.303
Calibrated stack	1320	0.739	0.054	0.143	0.214	0.000	2.493
Old EBM	1320	0.639	0.061	0.193	0.222	0.000	2.522
EBM v5	1320	0.667	0.058	0.204	0.222	0.000	2.614
Oracle grid	1320	0.771	0.036	0.069	0.159	0.000	2.691

Table 2: Hostile split success rates. These are the splits most likely to falsify compositional grasping claims.

Split	FC	Coll-FC	CEM	Robust	MLP	Stack	Old	v5	Oracle
combined_composition_shift	0.067	0.050	0.042	0.042	0.050	0.075	0.050	0.067	0.225
long_bar_high_aspect	0.283	0.050	0.325	0.033	0.900	0.975	0.842	0.883	0.900
thin_handle_shift	0.892	0.900	0.883	0.867	0.975	0.992	0.917	0.975	1.000
adversarial_clutter_gap	0.000	0.008	0.000	0.000	0.000	0.000	0.017	0.008	0.067
material_shift_proxy	0.250	0.308	0.250	0.242	0.467	0.692	0.408	0.492	0.683

The reason macro records the frozen explanation: v5 does not beat strongest non-oracle aggregate baseline ensemble_uncertainty_ranker by 0.030; v5 does not beat strongest combined-shift baseline calibrated_stacked_ranker by 0.030; ablation gate fails because monolithic_scalar_energy_only, no_cem_fallback, no_collision_energy matches or beats full v5; maximum stress gate fails against ensemble_uncertainty_ranker.

Table 2 isolates the hostile splits. These rows are where a compositional method should matter most. Long bars and thin handles test object/task decomposition. Adversarial clutter tests collision and feasibility energies. Material shift tests contact and robust perturbation energy. If v5 only matches force closure or CEM on these splits, the compositional mechanism is not established.

7 ABLATIONS

The ablation table is a primary result, not an appendix decoration. The mechanism claim requires that object, contact, task, collision, feasibility, robust perturbation, learned context, calibrated stacking, hard collision filtering, and fallback each have a reason to exist. If ablations match full v5, then the paper cannot attribute performance to the claimed decomposition. This is especially important because the hard fallback can mask the internal energy terms.

8 STRESS SWEEP

The stress sweep jointly increases friction difficulty, clutter tightness, aspect ratio, sensor noise, and task constraint strength. A robust compositional score should degrade gracefully and remain on the success/safety frontier. The maximum-stress row is a frozen decision gate. The oracle is included only to show whether the candidate set contains any recoverable action.

9 INTERPRETATION

The result should not be read as “energy-based grasping is impossible.” It should be read as: this specific compositional EBM/Transformer mechanism is not proven by the available evidence. The strongest pressure comes from three places. First, analytic baselines are much stronger than they look

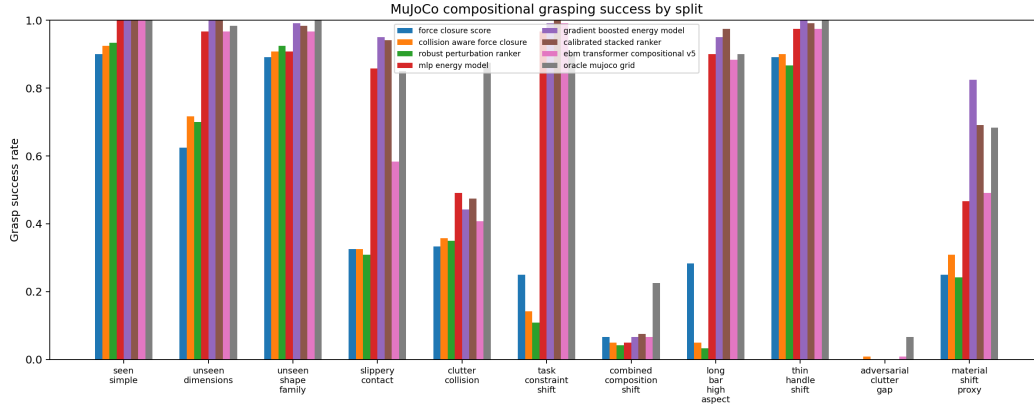


Figure 1: Frozen MuJoCo success by split. The figure includes strong analytic, learned, robust, stacked, v5, and oracle baselines.



Figure 2: Collision plus unsafe-force failures. A method that gains success by accepting unsafe grasps would not pass the gate.

when they include collision and robust perturbation terms. Second, learned scalar and stacked rankers often capture enough of the same structure without requiring the paper’s factorization story. Third, the fallback that improves v5 also weakens the claim that object/contact/task/collision/feasibility energies are independently necessary.

This matters for submission readiness. An ICLR reviewer does not need to disprove the idea in general. They only need to show that the submitted evidence does not support the claimed mechanism. The frozen ablation and stress gates are designed to make that critique visible before submission.

10 LIMITATIONS

This paper has substantial limitations even after expansion. It is a MuJoCo study, not hardware. It uses abstract object and candidate features, not raw RGB-D perception. It does not evaluate on Dex-Net, GraspNet, or Contact-GraspNet benchmarks. It does not train a large visual model or collect real grasp attempts. The Transformer is deliberately small to keep the CPU-only protocol tractable. The oracle evaluates a limited top-k set rather than an exhaustive continuous grasp space.

These limitations would be acceptable for a strong-revise local-mechanism paper only if the method clearly beat the local baselines. It does not. Therefore the limitations are not footnotes; they are part of the terminal decision.

Table 3: Aggregate hostile ablations over the three pre-registered hostile ablation splits.

Variant	Episodes	Success	Drop	Collision	Energy
Scalar only	360	0.239	0.478	0.617	2.855
No contact	360	0.225	0.508	0.625	2.965
No robust	360	0.203	0.500	0.617	3.013
No object	360	0.192	0.539	0.619	2.973
Full v5	360	0.189	0.544	0.619	2.954
No fallback	360	0.189	0.544	0.619	2.964
No feasibility	360	0.189	0.542	0.617	2.966
No hard filter	360	0.189	0.544	0.619	2.961
No task	360	0.189	0.547	0.625	2.948
No collision	360	0.186	0.544	0.625	2.971
No transformer	360	0.183	0.536	0.622	2.974
Old v4	360	0.158	0.561	0.611	2.945
No stack	360	0.133	0.600	0.619	2.900

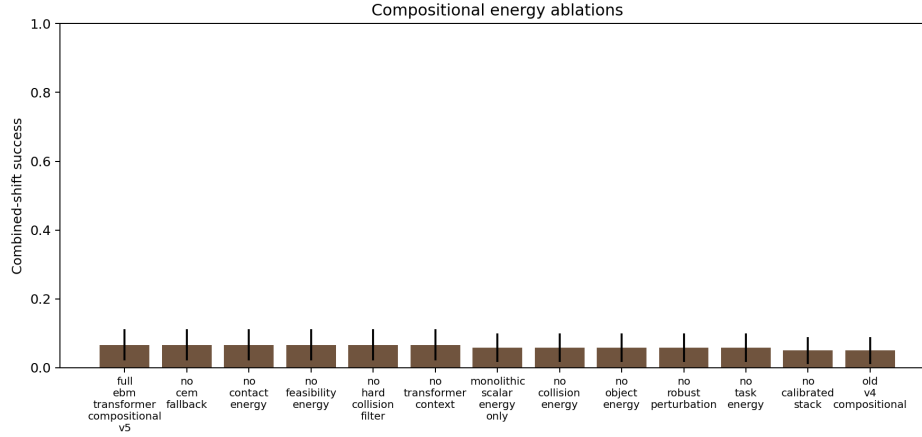


Figure 3: Hostile ablation success. The plot is allowed to embarrass the method; that is the point of the audit.

11 NEGATIVE CASES

The negative cases are not evaluated as full experiments in this paper. They are pre-specified boundaries showing where the current benchmark would overclaim. Deformable geometry, occluded shape, adversarial material, and semantic task ambiguity all require evidence beyond this local feature-based MuJoCo setup.

12 REPRODUCIBILITY

The final PDF is generated from CSV artifacts. The build script first renders paper/generated/*.tex, then compiles the manuscript, then copies the PDF to C:/Users/wangz/Downloads/68.pdf. No visible Desktop PDF is created. The validator checks Python compilation, row counts, figure presence, bright boxed citation settings, generated table inclusion, the Downloads PDF, page count, and Desktop hygiene.

13 TERMINAL DECISION

The terminal decision is **KILL_ARCHIVE**. The frozen reason is:

v5 does not beat strongest non-oracle aggregate baseline ensemble_uncertainty_ranker by 0.030; v5 does not beat strongest combined-shift baseline calibrated_stacked_ranker by 0.030; ablation gate fails because mono-

Table 4: Stress sweep success rates. Stress jointly changes friction, clutter clearance, aspect ratio, sensor noise, and task constraint strength.

Method	0.0	0.2	0.4	0.6	0.8	1.0
CEM analytic	0.912	0.900	0.812	0.138	0.037	0.000
Collision-aware FC	0.875	0.900	0.800	0.188	0.037	0.000
Robust perturb.	0.825	0.825	0.787	0.175	0.037	0.000
MLP energy	1.000	1.000	0.900	0.100	0.000	0.000
HGB energy	1.000	1.000	1.000	0.138	0.062	0.000
Calibrated stack	1.000	1.000	0.975	0.200	0.025	0.000
EBM v5	0.988	1.000	0.887	0.200	0.050	0.000
Oracle grid	1.000	1.000	1.000	0.650	0.212	0.025

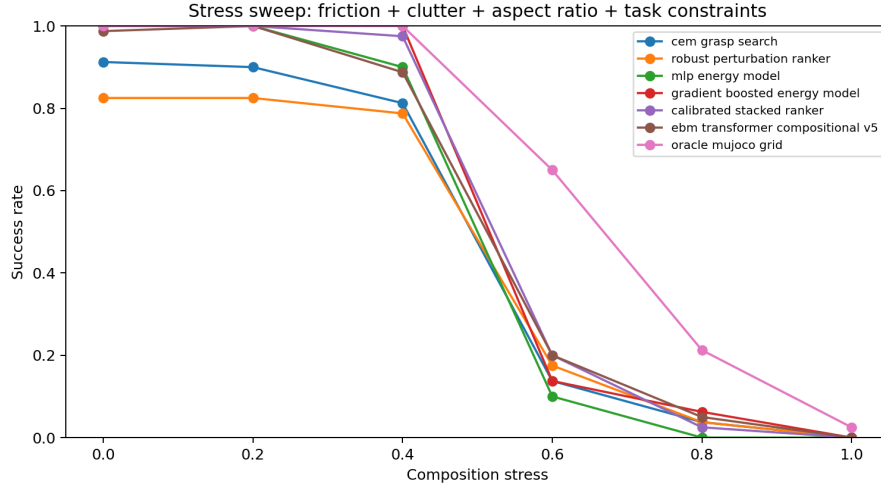


Figure 4: Stress sweep over composition difficulty. The maximum-stress comparison is pre-registered as a terminal gate.

lithic_scalar_energy_only, no_cem_fallback, no_collision_energy matches or beats full v5; maximum stress gate fails against ensemble_uncertainty_ranker

This is an archive result. The paper is valuable as a record of a plausible mechanism failing under stronger evidence. It should not be submitted as a positive ICLR-main paper in its current form.

A FULL MAIN METRICS

Table 6: Full split-level main metrics.

Split	Method	Episodes	Success	Slip	Drop	Collision	Energy
adversarial_clutter_gap	Random	120	0.008	0.088	0.842	0.992	1.966
adversarial_clutter_gap	Antipodal	120	0.000	0.088	0.775	0.983	2.671
adversarial_clutter_gap	Force closure	120	0.000	0.080	0.733	0.992	2.849
adversarial_clutter_gap	Collision-aware FC	120	0.008	0.082	0.717	0.975	3.157
adversarial_clutter_gap	CEM analytic	120	0.000	0.084	0.692	0.975	3.156
adversarial_clutter_gap	Robust perturb.	120	0.000	0.086	0.733	0.992	3.054
adversarial_clutter_gap	MLP energy	120	0.000	0.088	0.817	1.000	2.385
adversarial_clutter_gap	HGB energy	120	0.000	0.089	0.750	0.975	2.431
adversarial_clutter_gap	Transformer	120	0.000	0.085	0.800	0.992	2.071
adversarial_clutter_gap	Ensemble	120	0.000	0.095	0.775	0.992	2.653
adversarial_clutter_gap	Calibrated stack	120	0.000	0.090	0.750	0.992	2.926
adversarial_clutter_gap	Old EBM	120	0.017	0.086	0.708	0.975	3.046
adversarial_clutter_gap	EBM v5	120	0.008	0.088	0.742	0.983	3.060
adversarial_clutter_gap	Oracle grid	120	0.067	0.051	0.233	0.900	4.259
clutter_collision	Random	120	0.092	0.064	0.458	0.908	1.772
clutter_collision	Antipodal	120	0.333	0.057	0.142	0.633	2.659
clutter_collision	Force closure	120	0.333	0.056	0.208	0.650	2.636
clutter_collision	Collision-aware FC	120	0.358	0.060	0.158	0.600	2.653

Split	Method	Episodes	Success	Slip	Drop	Collision	Energy
clutter_collision	CEM analytic	120	0.358	0.059	0.150	0.600	2.685
clutter_collision	Robust perturb.	120	0.350	0.061	0.158	0.608	2.652
clutter_collision	MLP energy	120	0.492	0.045	0.158	0.500	2.314
clutter_collision	HGB energy	120	0.442	0.049	0.133	0.542	2.396
clutter_collision	Transformer	120	0.400	0.058	0.100	0.575	2.600
clutter_collision	Ensemble	120	0.417	0.056	0.142	0.542	2.388
clutter_collision	Calibrated stack	120	0.475	0.047	0.133	0.508	2.442
clutter_collision	Old EBM	120	0.383	0.057	0.117	0.592	2.638
clutter_collision	EBM v5	120	0.408	0.054	0.133	0.575	2.588
clutter_collision	Oracle grid	120	0.875	0.035	0.000	0.117	2.604
combined_composition_shift	Random	120	0.008	0.084	0.650	0.983	2.218
combined_composition_shift	Antipodal	120	0.050	0.093	0.400	0.867	3.642
combined_composition_shift	Force closure	120	0.067	0.088	0.400	0.875	3.656
combined_composition_shift	Collision-aware FC	120	0.050	0.089	0.425	0.900	3.469
combined_composition_shift	CEM analytic	120	0.042	0.089	0.383	0.900	3.719
combined_composition_shift	Robust perturb.	120	0.042	0.088	0.433	0.892	3.392
combined_composition_shift	MLP energy	120	0.050	0.073	0.483	0.950	2.851
combined_composition_shift	HGB energy	120	0.067	0.087	0.492	0.867	3.223
combined_composition_shift	Transformer	120	0.025	0.074	0.508	0.933	2.740
combined_composition_shift	Ensemble	120	0.075	0.086	0.500	0.850	3.330
combined_composition_shift	Calibrated stack	120	0.075	0.080	0.367	0.858	3.579
combined_composition_shift	Old EBM	120	0.050	0.089	0.408	0.875	3.663
combined_composition_shift	EBM v5	120	0.067	0.085	0.417	0.883	3.493
combined_composition_shift	Oracle grid	120	0.225	0.045	0.033	0.733	4.529
long_bar_high_aspect	Random	120	0.150	0.097	0.792	0.000	1.240
long_bar_high_aspect	Antipodal	120	0.183	0.068	0.758	0.000	0.749
long_bar_high_aspect	Force closure	120	0.283	0.062	0.692	0.000	0.807
long_bar_high_aspect	Collision-aware FC	120	0.050	0.063	0.950	0.000	0.623
long_bar_high_aspect	CEM analytic	120	0.325	0.062	0.642	0.000	0.819
long_bar_high_aspect	Robust perturb.	120	0.033	0.066	0.967	0.000	0.611
long_bar_high_aspect	MLP energy	120	0.900	0.057	0.008	0.000	1.467
long_bar_high_aspect	HGB energy	120	0.950	0.046	0.050	0.000	1.374
long_bar_high_aspect	Transformer	120	0.858	0.057	0.017	0.000	1.566
long_bar_high_aspect	Ensemble	120	0.958	0.050	0.042	0.000	1.378
long_bar_high_aspect	Calibrated stack	120	0.975	0.051	0.017	0.000	1.697
long_bar_high_aspect	Old EBM	120	0.842	0.056	0.025	0.000	1.633
long_bar_high_aspect	EBM v5	120	0.883	0.055	0.050	0.000	1.692
long_bar_high_aspect	Oracle grid	120	0.900	0.041	0.058	0.000	1.441
material_shift_proxy	Random	120	0.167	0.091	0.800	0.000	1.546
material_shift_proxy	Antipodal	120	0.325	0.087	0.583	0.000	2.119
material_shift_proxy	Force closure	120	0.250	0.080	0.675	0.000	2.020
material_shift_proxy	Collision-aware FC	120	0.308	0.075	0.633	0.000	2.088
material_shift_proxy	CEM analytic	120	0.250	0.081	0.683	0.000	2.011
material_shift_proxy	Robust perturb.	120	0.242	0.078	0.725	0.000	2.029
material_shift_proxy	MLP energy	120	0.467	0.072	0.483	0.000	1.906
material_shift_proxy	HGB energy	120	0.825	0.059	0.125	0.000	2.007
material_shift_proxy	Transformer	120	0.558	0.061	0.417	0.000	2.071
material_shift_proxy	Ensemble	120	0.858	0.057	0.067	0.000	2.033
material_shift_proxy	Calibrated stack	120	0.692	0.064	0.250	0.000	2.192
material_shift_proxy	Old EBM	120	0.408	0.065	0.567	0.000	2.132
material_shift_proxy	EBM v5	120	0.492	0.066	0.492	0.000	2.311
material_shift_proxy	Oracle grid	120	0.683	0.055	0.242	0.000	2.480
seen_simple	Random	120	0.867	0.065	0.017	0.000	2.030
seen_simple	Antipodal	120	0.808	0.069	0.000	0.000	2.370
seen_simple	Force closure	120	0.900	0.055	0.000	0.000	2.401
seen_simple	Collision-aware FC	120	0.925	0.054	0.000	0.000	2.671
seen_simple	CEM analytic	120	0.875	0.056	0.000	0.000	2.575
seen_simple	Robust perturb.	120	0.933	0.056	0.000	0.000	2.771
seen_simple	MLP energy	120	1.000	0.029	0.000	0.000	2.143
seen_simple	HGB energy	120	1.000	0.034	0.000	0.000	2.481
seen_simple	Transformer	120	0.983	0.035	0.000	0.000	2.112
seen_simple	Ensemble	120	1.000	0.041	0.000	0.000	2.529
seen_simple	Calibrated stack	120	1.000	0.035	0.000	0.000	2.758
seen_simple	Old EBM	120	0.983	0.046	0.000	0.000	2.592
seen_simple	EBM v5	120	1.000	0.041	0.000	0.000	2.753
seen_simple	Oracle grid	120	1.000	0.015	0.000	0.000	2.565
slippery_contact	Random	120	0.417	0.087	0.525	0.000	1.746
slippery_contact	Antipodal	120	0.375	0.092	0.508	0.000	2.082
slippery_contact	Force closure	120	0.325	0.081	0.625	0.000	1.985
slippery_contact	Collision-aware FC	120	0.325	0.080	0.642	0.000	2.047
slippery_contact	CEM analytic	120	0.308	0.082	0.633	0.000	2.027
slippery_contact	Robust perturb.	120	0.308	0.083	0.658	0.000	2.058
slippery_contact	MLP energy	120	0.858	0.054	0.125	0.000	1.953
slippery_contact	HGB energy	120	0.950	0.054	0.033	0.000	2.079
slippery_contact	Transformer	120	0.883	0.052	0.100	0.000	1.929
slippery_contact	Ensemble	120	0.933	0.059	0.025	0.000	1.988
slippery_contact	Calibrated stack	120	0.942	0.052	0.058	0.000	2.097
slippery_contact	Old EBM	120	0.683	0.059	0.275	0.000	2.102
slippery_contact	EBM v5	120	0.583	0.066	0.392	0.000	2.169

Split	Method	Episodes	Success	Slip	Drop	Collision	Energy
slippery_contact	Oracle grid	120	0.850	0.053	0.117	0.000	2.508
task_constraint_shift	Random	120	0.183	0.073	0.650	0.000	1.649
task_constraint_shift	Antipodal	120	0.067	0.064	0.675	0.000	2.010
task_constraint_shift	Force closure	120	0.250	0.063	0.725	0.000	0.837
task_constraint_shift	Collision-aware FC	120	0.142	0.065	0.858	0.000	0.774
task_constraint_shift	CEM analytic	120	0.308	0.062	0.658	0.000	0.922
task_constraint_shift	Robust perturb.	120	0.108	0.068	0.892	0.000	0.741
task_constraint_shift	MLP energy	120	0.967	0.040	0.000	0.000	1.596
task_constraint_shift	HGB energy	120	0.992	0.040	0.008	0.000	1.575
task_constraint_shift	Transformer	120	0.933	0.045	0.000	0.000	1.707
task_constraint_shift	Ensemble	120	1.000	0.047	0.000	0.000	1.625
task_constraint_shift	Calibrated stack	120	1.000	0.047	0.000	0.000	1.838
task_constraint_shift	Old EBM	120	0.942	0.053	0.000	0.000	1.711
task_constraint_shift	EBM v5	120	0.992	0.049	0.000	0.000	1.751
task_constraint_shift	Oracle grid	120	0.900	0.040	0.067	0.000	1.618
thin_handle_shift	Random	120	0.342	0.077	0.533	0.000	1.368
thin_handle_shift	Antipodal	120	0.000	0.075	0.333	0.000	3.222
thin_handle_shift	Force closure	120	0.892	0.053	0.008	0.000	2.031
thin_handle_shift	Collision-aware FC	120	0.900	0.052	0.000	0.000	2.145
thin_handle_shift	CEM analytic	120	0.883	0.055	0.000	0.000	2.167
thin_handle_shift	Robust perturb.	120	0.867	0.056	0.000	0.000	2.197
thin_handle_shift	MLP energy	120	0.975	0.044	0.000	0.000	1.834
thin_handle_shift	HGB energy	120	1.000	0.044	0.000	0.000	2.036
thin_handle_shift	Transformer	120	0.933	0.047	0.000	0.000	2.081
thin_handle_shift	Ensemble	120	0.992	0.046	0.000	0.000	2.131
thin_handle_shift	Calibrated stack	120	0.992	0.044	0.000	0.000	2.188
thin_handle_shift	Old EBM	120	0.917	0.053	0.000	0.000	2.201
thin_handle_shift	EBM v5	120	0.975	0.047	0.000	0.000	2.204
thin_handle_shift	Oracle grid	120	1.000	0.020	0.000	0.000	1.627
unseen_dimensions	Random	120	0.392	0.069	0.550	0.000	1.553
unseen_dimensions	Antipodal	120	0.733	0.063	0.092	0.000	4.109
unseen_dimensions	Force closure	120	0.625	0.064	0.208	0.000	3.814
unseen_dimensions	Collision-aware FC	120	0.717	0.064	0.133	0.000	4.028
unseen_dimensions	CEM analytic	120	0.642	0.064	0.208	0.000	3.790
unseen_dimensions	Robust perturb.	120	0.700	0.065	0.233	0.000	3.663
unseen_dimensions	MLP energy	120	0.967	0.038	0.008	0.000	2.185
unseen_dimensions	HGB energy	120	1.000	0.043	0.000	0.000	2.075
unseen_dimensions	Transformer	120	0.975	0.039	0.000	0.000	2.062
unseen_dimensions	Ensemble	120	1.000	0.046	0.000	0.000	1.944
unseen_dimensions	Calibrated stack	120	1.000	0.041	0.000	0.000	2.155
unseen_dimensions	Old EBM	120	0.900	0.049	0.025	0.000	2.420
unseen_dimensions	EBM v5	120	0.967	0.039	0.017	0.000	3.094
unseen_dimensions	Oracle grid	120	0.983	0.025	0.008	0.000	3.103
unseen_shape_family	Random	120	0.392	0.067	0.067	0.000	2.534
unseen_shape_family	Antipodal	120	0.367	0.059	0.000	0.000	2.995
unseen_shape_family	Force closure	120	0.892	0.058	0.000	0.000	3.474
unseen_shape_family	Collision-aware FC	120	0.908	0.060	0.000	0.000	3.616
unseen_shape_family	CEM analytic	120	0.925	0.059	0.000	0.000	3.611
unseen_shape_family	Robust perturb.	120	0.925	0.060	0.000	0.000	3.713
unseen_shape_family	MLP energy	120	0.908	0.037	0.000	0.000	3.243
unseen_shape_family	HGB energy	120	0.992	0.049	0.000	0.000	3.277
unseen_shape_family	Transformer	120	0.917	0.054	0.000	0.000	3.519
unseen_shape_family	Ensemble	120	0.983	0.055	0.000	0.000	3.331
unseen_shape_family	Calibrated stack	120	0.983	0.048	0.000	0.000	3.551
unseen_shape_family	Old EBM	120	0.900	0.057	0.000	0.000	3.605
unseen_shape_family	EBM v5	120	0.967	0.051	0.000	0.000	3.635
unseen_shape_family	Oracle grid	120	1.000	0.016	0.000	0.000	2.864

B FULL ABLATION METRICS

Table 7: Full hostile split ablation metrics.

Split	Method	Episodes	Success	Slip	Drop	Collision	Energy
adversarial_clutter_gap	Full v5	120	0.008	0.087	0.733	0.983	3.063
adversarial_clutter_gap	Scalar only	120	0.008	0.096	0.733	0.983	2.703
adversarial_clutter_gap	No stack	120	0.008	0.089	0.750	0.983	3.027
adversarial_clutter_gap	No fallback	120	0.008	0.086	0.733	0.983	3.079
adversarial_clutter_gap	No collision	120	0.008	0.085	0.733	0.983	3.085
adversarial_clutter_gap	No contact	120	0.008	0.086	0.733	0.983	3.084
adversarial_clutter_gap	No feasibility	120	0.008	0.086	0.733	0.975	3.085
adversarial_clutter_gap	No hard filter	120	0.008	0.087	0.733	0.983	3.071
adversarial_clutter_gap	No object	120	0.008	0.087	0.733	0.983	3.056
adversarial_clutter_gap	No robust	120	0.008	0.087	0.683	0.975	3.102
adversarial_clutter_gap	No task	120	0.008	0.087	0.750	0.983	3.058

Split	Method	Episodes	Success	Slip	Drop	Collision	Energy
adversarial_clutter_gap	No transformer	120	0.008	0.086	0.725	0.983	3.066
adversarial_clutter_gap	Old v4	120	0.017	0.086	0.708	0.967	3.067
combined_composition_shift	Full v5	120	0.067	0.086	0.408	0.875	3.489
combined_composition_shift	Scalar only	120	0.058	0.086	0.442	0.867	3.571
combined_composition_shift	No stack	120	0.050	0.086	0.408	0.875	3.533
combined_composition_shift	No fallback	120	0.067	0.085	0.408	0.875	3.500
combined_composition_shift	No collision	120	0.058	0.084	0.408	0.892	3.516
combined_composition_shift	No contact	120	0.067	0.086	0.408	0.892	3.518
combined_composition_shift	No feasibility	120	0.067	0.085	0.400	0.875	3.511
combined_composition_shift	No hard filter	120	0.067	0.085	0.408	0.875	3.499
combined_composition_shift	No object	120	0.058	0.085	0.400	0.875	3.537
combined_composition_shift	No robust	120	0.058	0.085	0.392	0.875	3.682
combined_composition_shift	No task	120	0.058	0.085	0.417	0.892	3.474
combined_composition_shift	No transformer	120	0.067	0.084	0.400	0.883	3.554
combined_composition_shift	Old v4	120	0.050	0.088	0.408	0.867	3.636
material_shift_proxy	Full v5	120	0.492	0.066	0.492	0.000	2.311
material_shift_proxy	Scalar only	120	0.650	0.071	0.258	0.000	2.290
material_shift_proxy	No stack	120	0.342	0.067	0.642	0.000	2.141
material_shift_proxy	No fallback	120	0.492	0.066	0.492	0.000	2.311
material_shift_proxy	No collision	120	0.492	0.066	0.492	0.000	2.311
material_shift_proxy	No contact	120	0.600	0.062	0.383	0.000	2.292
material_shift_proxy	No feasibility	120	0.492	0.065	0.492	0.000	2.302
material_shift_proxy	No hard filter	120	0.492	0.066	0.492	0.000	2.311
material_shift_proxy	No object	120	0.508	0.065	0.483	0.000	2.326
material_shift_proxy	No robust	120	0.542	0.066	0.425	0.000	2.256
material_shift_proxy	No task	120	0.500	0.066	0.475	0.000	2.311
material_shift_proxy	No transformer	120	0.475	0.068	0.483	0.000	2.304
material_shift_proxy	Old v4	120	0.408	0.065	0.567	0.000	2.132

C FULL STRESS METRICS

Table 8: Full stress-sweep metrics.

Split	Method	Episodes	Success	Slip	Drop	Collision	Energy
stress_0.00	Collision-aware FC	80	0.875	0.052	0.000	0.000	2.501
stress_0.00	CEM analytic	80	0.912	0.047	0.000	0.000	2.461
stress_0.00	Robust perturb.	80	0.825	0.059	0.000	0.000	2.573
stress_0.00	MLP energy	80	1.000	0.034	0.000	0.000	2.292
stress_0.00	HGB energy	80	1.000	0.038	0.000	0.000	2.441
stress_0.00	Ensemble	80	1.000	0.038	0.000	0.000	2.497
stress_0.00	Calibrated stack	80	1.000	0.036	0.000	0.000	2.684
stress_0.00	EBM v5	80	0.988	0.039	0.000	0.000	2.588
stress_0.00	Oracle grid	80	1.000	0.014	0.000	0.000	2.461
stress_0.20	Collision-aware FC	80	0.900	0.053	0.013	0.000	3.789
stress_0.20	CEM analytic	80	0.900	0.051	0.000	0.000	3.650
stress_0.20	Robust perturb.	80	0.825	0.063	0.037	0.000	3.664
stress_0.20	MLP energy	80	1.000	0.024	0.000	0.000	2.776
stress_0.20	HGB energy	80	1.000	0.047	0.000	0.000	2.368
stress_0.20	Ensemble	80	1.000	0.048	0.000	0.000	2.439
stress_0.20	Calibrated stack	80	1.000	0.042	0.000	0.000	2.528
stress_0.20	EBM v5	80	1.000	0.039	0.000	0.000	3.183
stress_0.20	Oracle grid	80	1.000	0.014	0.000	0.000	2.575
stress_0.40	Collision-aware FC	80	0.800	0.060	0.113	0.138	5.880
stress_0.40	CEM analytic	80	0.812	0.060	0.138	0.125	5.734
stress_0.40	Robust perturb.	80	0.787	0.061	0.125	0.175	5.761
stress_0.40	MLP energy	80	0.900	0.051	0.000	0.700	2.499
stress_0.40	HGB energy	80	1.000	0.051	0.000	0.713	2.365
stress_0.40	Ensemble	80	0.988	0.054	0.000	0.637	2.413
stress_0.40	Calibrated stack	80	0.975	0.049	0.013	0.575	2.641
stress_0.40	EBM v5	80	0.887	0.053	0.100	0.113	4.801
stress_0.40	Oracle grid	80	1.000	0.018	0.000	0.000	5.401
stress_0.60	Collision-aware FC	80	0.188	0.096	0.525	0.625	3.101
stress_0.60	CEM analytic	80	0.138	0.094	0.588	0.725	2.732
stress_0.60	Robust perturb.	80	0.175	0.084	0.537	0.625	3.265
stress_0.60	MLP energy	80	0.100	0.075	0.200	0.825	3.098
stress_0.60	HGB energy	80	0.138	0.075	0.400	0.738	2.447
stress_0.60	Ensemble	80	0.113	0.103	0.338	0.787	3.421
stress_0.60	Calibrated stack	80	0.200	0.083	0.450	0.675	3.459
stress_0.60	EBM v5	80	0.200	0.080	0.500	0.650	3.170
stress_0.60	Oracle grid	80	0.650	0.044	0.000	0.287	3.855
stress_0.80	Collision-aware FC	80	0.037	0.077	0.762	0.950	1.411
stress_0.80	CEM analytic	80	0.037	0.078	0.787	0.938	1.388
stress_0.80	Robust perturb.	80	0.037	0.076	0.762	0.938	1.457
stress_0.80	MLP energy	80	0.000	0.073	0.312	0.975	2.236

Split	Method	Episodes	Success	Slip	Drop	Collision	Energy
stress_0.80	HGB energy	80	0.062	0.086	0.500	0.875	2.633
stress_0.80	Ensemble	80	0.025	0.105	0.450	0.838	3.378
stress_0.80	Calibrated stack	80	0.025	0.080	0.700	0.900	1.900
stress_0.80	EBM v5	80	0.050	0.074	0.750	0.925	1.486
stress_0.80	Oracle grid	80	0.212	0.054	0.062	0.700	2.732
stress_1.00	Collision-aware FC	80	0.000	0.077	0.975	1.000	0.941
stress_1.00	CEM analytic	80	0.000	0.078	0.975	1.000	0.958
stress_1.00	Robust perturb.	80	0.000	0.075	0.950	1.000	0.985
stress_1.00	MLP energy	80	0.000	0.076	0.700	1.000	1.632
stress_1.00	HGB energy	80	0.000	0.105	0.650	1.000	2.377
stress_1.00	Ensemble	80	0.025	0.133	0.600	0.963	3.133
stress_1.00	Calibrated stack	80	0.000	0.073	0.887	1.000	1.153
stress_1.00	EBM v5	80	0.000	0.076	0.950	1.000	0.976
stress_1.00	Oracle grid	80	0.025	0.058	0.300	0.950	2.236

D ALL PAIRED COMPARISONS

Table 9: All split-level paired seed comparisons versus EBM v5. Positive values favor EBM v5.

Split	Baseline	Diff	t approx.	p approx.
seen_simple	Antipodal	0.192	8.205	0.000
seen_simple	Calibrated stack	0.000	0.000	1.000
seen_simple	CEM analytic	0.125	6.357	0.000
seen_simple	Collision-aware FC	0.075	3.211	0.001
seen_simple	Old EBM	0.017	1.528	0.127
seen_simple	Ensemble	0.000	0.000	1.000
seen_simple	Force closure	0.100	7.945	0.000
seen_simple	HGB energy	0.000	0.000	1.000
seen_simple	MLP energy	0.000	0.000	1.000
seen_simple	Oracle grid	0.000	0.000	1.000
seen_simple	Random	0.133	3.055	0.002
seen_simple	Robust perturb.	0.067	2.646	0.008
seen_simple	Transformer	0.017	1.528	0.127
unseen_dimensions	Antipodal	0.233	6.549	0.000
unseen_dimensions	Calibrated stack	-0.033	-1.871	0.061
unseen_dimensions	CEM analytic	0.325	8.882	0.000
unseen_dimensions	Collision-aware FC	0.250	7.127	0.000
unseen_dimensions	Old EBM	0.067	3.055	0.002
unseen_dimensions	Ensemble	-0.033	-1.871	0.061
unseen_dimensions	Force closure	0.342	11.632	0.000
unseen_dimensions	HGB energy	-0.033	-1.871	0.061
unseen_dimensions	MLP energy	0.000	0.000	1.000
unseen_dimensions	Oracle grid	-0.017	-1.000	0.317
unseen_dimensions	Random	0.575	22.996	0.000
unseen_dimensions	Robust perturb.	0.267	7.056	0.000
unseen_dimensions	Transformer	-0.008	-0.423	0.672
unseen_shape_family	Antipodal	0.600	13.749	0.000
unseen_shape_family	Calibrated stack	-0.017	-1.528	0.127
unseen_shape_family	CEM analytic	0.042	2.376	0.018
unseen_shape_family	Collision-aware FC	0.058	3.861	0.000
unseen_shape_family	Old EBM	0.067	2.646	0.008
unseen_shape_family	Ensemble	-0.017	-1.528	0.127
unseen_shape_family	Force closure	0.075	4.964	0.000
unseen_shape_family	HGB energy	-0.025	-2.049	0.040
unseen_shape_family	MLP energy	0.058	2.199	0.028
unseen_shape_family	Oracle grid	-0.033	-1.871	0.061
unseen_shape_family	Random	0.575	13.799	0.000
unseen_shape_family	Robust perturb.	0.042	2.377	0.018
unseen_shape_family	Transformer	0.050	2.049	0.040
slippery_contact	Antipodal	0.208	3.490	0.001
slippery_contact	Calibrated stack	-0.358	-10.093	0.000
slippery_contact	CEM analytic	0.275	4.093	0.000
slippery_contact	Collision-aware FC	0.258	8.080	0.000
slippery_contact	Old EBM	-0.100	-2.122	0.034
slippery_contact	Ensemble	-0.350	-8.896	0.000
slippery_contact	Force closure	0.258	5.395	0.000
slippery_contact	HGB energy	-0.367	-13.012	0.000
slippery_contact	MLP energy	-0.275	-4.951	0.000
slippery_contact	Oracle grid	-0.267	-5.465	0.000
slippery_contact	Random	0.167	2.645	0.008
slippery_contact	Robust perturb.	0.275	5.227	0.000
slippery_contact	Transformer	-0.300	-7.937	0.000
clutter_collision	Antipodal	0.075	2.553	0.011
clutter_collision	Calibrated stack	-0.067	-2.000	0.045

Split	Baseline	Diff	t approx.	p approx.
clutter_collision	CEM analytic	0.050	1.210	0.226
clutter_collision	Collision-aware FC	0.050	1.527	0.127
clutter_collision	Old EBM	0.025	1.158	0.247
clutter_collision	Ensemble	-0.008	-0.215	0.830
clutter_collision	Force closure	0.075	2.346	0.019
clutter_collision	HGB energy	-0.033	-0.624	0.533
clutter_collision	MLP energy	-0.083	-3.988	0.000
clutter_collision	Oracle grid	-0.467	-13.096	0.000
clutter_collision	Random	0.317	7.667	0.000
clutter_collision	Robust perturb.	0.058	1.508	0.132
clutter_collision	Transformer	0.008	0.315	0.753
task_constraint_shift	Antipodal	0.925	61.261	0.000
task_constraint_shift	Calibrated stack	-0.008	-1.000	0.317
task_constraint_shift	CEM analytic	0.683	24.887	0.000
task_constraint_shift	Collision-aware FC	0.850	24.233	0.000
task_constraint_shift	Old EBM	0.050	3.000	0.003
task_constraint_shift	Ensemble	-0.008	-1.000	0.317
task_constraint_shift	Force closure	0.742	23.199	0.000
task_constraint_shift	HGB energy	0.000	0.000	1.000
task_constraint_shift	MLP energy	0.025	0.893	0.372
task_constraint_shift	Oracle grid	0.092	4.247	0.000
task_constraint_shift	Random	0.808	27.514	0.000
task_constraint_shift	Robust perturb.	0.883	26.986	0.000
task_constraint_shift	Transformer	0.058	2.498	0.013
combined_composition_shift	Antipodal	0.017	1.528	0.127
combined_composition_shift	Calibrated stack	-0.008	-1.000	0.317
combined_composition_shift	CEM analytic	0.025	2.049	0.040
combined_composition_shift	Collision-aware FC	0.017	1.528	0.127
combined_composition_shift	Old EBM	0.017	1.000	0.317
combined_composition_shift	Ensemble	-0.008	-0.552	0.581
combined_composition_shift	Force closure	0.000	0.000	1.000
combined_composition_shift	HGB energy	0.000	0.001	1.000
combined_composition_shift	MLP energy	0.017	1.528	0.127
combined_composition_shift	Oracle grid	-0.158	-5.655	0.000
combined_composition_shift	Random	0.058	2.966	0.003
combined_composition_shift	Robust perturb.	0.025	2.049	0.040
combined_composition_shift	Transformer	0.042	2.376	0.018
long_bar_high_aspect	Antipodal	0.700	12.745	0.000
long_bar_high_aspect	Calibrated stack	-0.092	-3.274	0.001
long_bar_high_aspect	CEM analytic	0.558	18.188	0.000
long_bar_high_aspect	Collision-aware FC	0.833	26.999	0.000
long_bar_high_aspect	Old EBM	0.042	1.666	0.096
long_bar_high_aspect	Ensemble	-0.075	-2.183	0.029
long_bar_high_aspect	Force closure	0.600	19.439	0.000
long_bar_high_aspect	HGB energy	-0.067	-1.999	0.046
long_bar_high_aspect	MLP energy	-0.017	-0.551	0.581
long_bar_high_aspect	Oracle grid	-0.017	-0.607	0.544
long_bar_high_aspect	Random	0.733	19.404	0.000
long_bar_high_aspect	Robust perturb.	0.850	28.135	0.000
long_bar_high_aspect	Transformer	0.025	0.664	0.507
thin_handle_shift	Antipodal	0.975	55.605	0.000
thin_handle_shift	Calibrated stack	-0.017	-1.528	0.127
thin_handle_shift	CEM analytic	0.092	2.986	0.003
thin_handle_shift	Collision-aware FC	0.075	2.346	0.019
thin_handle_shift	Old EBM	0.058	3.863	0.000
thin_handle_shift	Ensemble	-0.017	-0.797	0.425
thin_handle_shift	Force closure	0.083	5.003	0.000
thin_handle_shift	HGB energy	-0.025	-1.426	0.154
thin_handle_shift	MLP energy	0.000	0.001	1.000
thin_handle_shift	Oracle grid	-0.025	-1.426	0.154
thin_handle_shift	Random	0.633	12.569	0.000
thin_handle_shift	Robust perturb.	0.108	3.870	0.000
thin_handle_shift	Transformer	0.042	1.931	0.053
adversarial_clutter_gap	Antipodal	0.008	1.000	0.317
adversarial_clutter_gap	Calibrated stack	0.008	1.000	0.317
adversarial_clutter_gap	CEM analytic	0.008	1.000	0.317
adversarial_clutter_gap	Collision-aware FC	0.000	0.000	1.000
adversarial_clutter_gap	Old EBM	-0.008	-1.000	0.317
adversarial_clutter_gap	Ensemble	0.008	1.000	0.317
adversarial_clutter_gap	Force closure	0.008	1.000	0.317
adversarial_clutter_gap	HGB energy	0.008	1.000	0.317
adversarial_clutter_gap	MLP energy	0.008	1.000	0.317
adversarial_clutter_gap	Oracle grid	-0.058	-2.497	0.013
adversarial_clutter_gap	Random	0.000	0.000	1.000
adversarial_clutter_gap	Robust perturb.	0.008	1.000	0.317
adversarial_clutter_gap	Transformer	0.008	1.000	0.317
material_shift_proxy	Antipodal	0.167	1.871	0.061
material_shift_proxy	Calibrated stack	-0.200	-3.642	0.000
material_shift_proxy	CEM analytic	0.242	3.074	0.002

Split	Baseline	Diff	t approx.	p approx.
material_shift_proxy	Collision-aware FC	0.183	2.583	0.010
material_shift_proxy	Old EBM	0.083	2.759	0.006
material_shift_proxy	Ensemble	-0.367	-6.351	0.000
material_shift_proxy	Force closure	0.242	4.193	0.000
material_shift_proxy	HGB energy	-0.333	-4.752	0.000
material_shift_proxy	MLP energy	0.025	0.424	0.672
material_shift_proxy	Oracle grid	-0.192	-2.593	0.009
material_shift_proxy	Random	0.325	4.755	0.000
material_shift_proxy	Robust perturb.	0.250	3.319	0.001
material_shift_proxy	Transformer	-0.067	-1.366	0.172

E SELECTED SEED METRICS

Table 10: Selected seed-level metrics for reproducibility.

Split	Method	Episodes	Success	Slip	Drop	Collision	Energy
adversarial_clutter_gap	CEM analytic	15	0.000	0.073	0.800	0.867	2.671
adversarial_clutter_gap	CEM analytic	15	0.000	0.064	0.400	1.000	3.529
adversarial_clutter_gap	CEM analytic	15	0.000	0.086	0.800	1.000	3.179
adversarial_clutter_gap	CEM analytic	15	0.000	0.111	0.800	0.933	2.575
adversarial_clutter_gap	CEM analytic	15	0.000	0.085	0.733	1.000	2.905
adversarial_clutter_gap	CEM analytic	15	0.000	0.100	0.533	1.000	4.426
adversarial_clutter_gap	CEM analytic	15	0.000	0.085	0.733	1.000	2.553
adversarial_clutter_gap	CEM analytic	15	0.000	0.064	0.733	1.000	3.406
adversarial_clutter_gap	Robust perturb.	15	0.000	0.075	0.733	0.933	3.204
adversarial_clutter_gap	Robust perturb.	15	0.000	0.076	0.467	1.000	3.337
adversarial_clutter_gap	Robust perturb.	15	0.000	0.090	0.800	1.000	2.720
adversarial_clutter_gap	Robust perturb.	15	0.000	0.094	0.867	1.000	2.987
adversarial_clutter_gap	Robust perturb.	15	0.000	0.083	0.733	1.000	2.800
adversarial_clutter_gap	Robust perturb.	15	0.000	0.103	0.733	1.000	3.600
adversarial_clutter_gap	Robust perturb.	15	0.000	0.093	0.800	1.000	2.411
adversarial_clutter_gap	Robust perturb.	15	0.000	0.075	0.733	1.000	3.372
adversarial_clutter_gap	EBM v5	15	0.000	0.081	0.800	0.933	3.130
adversarial_clutter_gap	EBM v5	15	0.000	0.079	0.533	1.000	3.253
adversarial_clutter_gap	EBM v5	15	0.000	0.091	0.733	1.000	2.921
adversarial_clutter_gap	EBM v5	15	0.000	0.094	0.867	1.000	2.977
adversarial_clutter_gap	EBM v5	15	0.000	0.082	0.733	1.000	2.848
adversarial_clutter_gap	EBM v5	15	0.067	0.105	0.667	0.933	3.801
adversarial_clutter_gap	EBM v5	15	0.000	0.092	0.800	1.000	2.458
adversarial_clutter_gap	EBM v5	15	0.000	0.077	0.800	1.000	3.095
adversarial_clutter_gap	Oracle grid	15	0.000	0.060	0.333	0.933	3.897
adversarial_clutter_gap	Oracle grid	15	0.000	0.047	0.067	1.000	4.209
adversarial_clutter_gap	Oracle grid	15	0.000	0.045	0.133	1.000	5.164
adversarial_clutter_gap	Oracle grid	15	0.133	0.071	0.200	0.800	4.358
adversarial_clutter_gap	Oracle grid	15	0.133	0.034	0.200	0.867	4.571
adversarial_clutter_gap	Oracle grid	15	0.133	0.039	0.133	0.867	4.439
adversarial_clutter_gap	Oracle grid	15	0.000	0.053	0.333	1.000	3.594
adversarial_clutter_gap	Oracle grid	15	0.133	0.057	0.467	0.733	3.837
clutter_collision	CEM analytic	15	0.333	0.083	0.133	0.533	2.565
clutter_collision	CEM analytic	15	0.267	0.059	0.133	0.733	2.755
clutter_collision	CEM analytic	15	0.400	0.058	0.067	0.600	2.709
clutter_collision	CEM analytic	15	0.200	0.064	0.400	0.733	2.107
clutter_collision	CEM analytic	15	0.200	0.053	0.067	0.800	2.864
clutter_collision	CEM analytic	15	0.467	0.058	0.133	0.400	2.850
clutter_collision	CEM analytic	15	0.600	0.039	0.133	0.400	2.801
clutter_collision	CEM analytic	15	0.400	0.057	0.133	0.600	2.826
clutter_collision	Robust perturb.	15	0.200	0.081	0.200	0.667	2.510
clutter_collision	Robust perturb.	15	0.267	0.069	0.133	0.733	2.673
clutter_collision	Robust perturb.	15	0.333	0.058	0.067	0.667	2.478
clutter_collision	Robust perturb.	15	0.200	0.064	0.333	0.733	2.235
clutter_collision	Robust perturb.	15	0.333	0.055	0.067	0.667	2.902
clutter_collision	Robust perturb.	15	0.533	0.055	0.133	0.333	2.850
clutter_collision	Robust perturb.	15	0.533	0.049	0.133	0.467	2.865
clutter_collision	Robust perturb.	15	0.400	0.062	0.200	0.600	2.704
clutter_collision	EBM v5	15	0.467	0.071	0.133	0.467	2.329
clutter_collision	EBM v5	15	0.267	0.061	0.133	0.733	2.674
clutter_collision	EBM v5	15	0.333	0.051	0.067	0.667	2.566
clutter_collision	EBM v5	15	0.333	0.058	0.267	0.600	2.292
clutter_collision	EBM v5	15	0.467	0.057	0.067	0.533	2.668
clutter_collision	EBM v5	15	0.467	0.039	0.067	0.533	2.806
clutter_collision	EBM v5	15	0.533	0.040	0.133	0.467	2.814
clutter_collision	EBM v5	15	0.400	0.058	0.200	0.600	2.559
clutter_collision	Oracle grid	15	0.800	0.029	0.000	0.200	2.673
clutter_collision	Oracle grid	15	0.933	0.033	0.000	0.067	2.562

Split	Method	Episodes	Success	Slip	Drop	Collision	Energy
clutter_collision	Oracle grid	15	0.733	0.047	0.000	0.200	2.775
clutter_collision	Oracle grid	15	0.867	0.046	0.000	0.133	2.643
clutter_collision	Oracle grid	15	0.933	0.044	0.000	0.067	2.344
clutter_collision	Oracle grid	15	0.933	0.029	0.000	0.067	2.563
clutter_collision	Oracle grid	15	0.933	0.019	0.000	0.067	2.729
clutter_collision	Oracle grid	15	0.867	0.031	0.000	0.133	2.544
combined_composition_shift	CEM analytic	15	0.000	0.106	0.400	0.867	4.055
combined_composition_shift	CEM analytic	15	0.000	0.105	0.333	1.000	3.650
combined_composition_shift	CEM analytic	15	0.067	0.072	0.133	0.867	4.875
combined_composition_shift	CEM analytic	15	0.067	0.089	0.467	0.867	3.389
combined_composition_shift	CEM analytic	15	0.000	0.105	0.533	0.933	3.038
combined_composition_shift	CEM analytic	15	0.067	0.078	0.333	0.867	3.754
combined_composition_shift	CEM analytic	15	0.067	0.070	0.400	0.933	3.859
combined_composition_shift	CEM analytic	15	0.067	0.088	0.467	0.867	3.130
combined_composition_shift	Robust perturb.	15	0.000	0.085	0.533	0.933	3.309
combined_composition_shift	Robust perturb.	15	0.000	0.104	0.400	0.933	3.535
combined_composition_shift	Robust perturb.	15	0.067	0.089	0.267	0.867	4.270
combined_composition_shift	Robust perturb.	15	0.067	0.087	0.400	0.800	3.232
combined_composition_shift	Robust perturb.	15	0.000	0.106	0.667	0.867	2.701
combined_composition_shift	Robust perturb.	15	0.067	0.084	0.467	0.867	3.453
combined_composition_shift	Robust perturb.	15	0.067	0.075	0.533	0.933	2.921
combined_composition_shift	Robust perturb.	15	0.067	0.075	0.200	0.933	3.718
combined_composition_shift	EBM v5	15	0.000	0.093	0.533	0.933	3.230
combined_composition_shift	EBM v5	15	0.000	0.087	0.400	1.000	3.384
combined_composition_shift	EBM v5	15	0.067	0.088	0.267	0.867	4.237
combined_composition_shift	EBM v5	15	0.133	0.080	0.267	0.800	3.836
combined_composition_shift	EBM v5	15	0.067	0.102	0.667	0.800	2.885
combined_composition_shift	EBM v5	15	0.067	0.089	0.467	0.867	3.448
combined_composition_shift	EBM v5	15	0.133	0.069	0.467	0.867	3.383
combined_composition_shift	EBM v5	15	0.067	0.073	0.267	0.933	3.543
combined_composition_shift	Oracle grid	15	0.133	0.045	0.000	0.800	5.614
combined_composition_shift	Oracle grid	15	0.067	0.056	0.067	0.867	3.785
combined_composition_shift	Oracle grid	15	0.333	0.039	0.000	0.667	4.812
combined_composition_shift	Oracle grid	15	0.333	0.036	0.000	0.667	4.981
combined_composition_shift	Oracle grid	15	0.133	0.078	0.133	0.733	3.191
combined_composition_shift	Oracle grid	15	0.200	0.044	0.000	0.800	4.732
combined_composition_shift	Oracle grid	15	0.400	0.035	0.067	0.533	5.101
combined_composition_shift	Oracle grid	15	0.200	0.032	0.000	0.800	4.020
long_bar_high_aspect	CEM analytic	15	0.400	0.060	0.533	0.000	0.774
long_bar_high_aspect	CEM analytic	15	0.333	0.068	0.667	0.000	0.805
long_bar_high_aspect	CEM analytic	15	0.333	0.067	0.600	0.000	0.806
long_bar_high_aspect	CEM analytic	15	0.267	0.069	0.667	0.000	0.836
long_bar_high_aspect	CEM analytic	15	0.267	0.064	0.733	0.000	0.865
long_bar_high_aspect	CEM analytic	15	0.267	0.057	0.733	0.000	0.747
long_bar_high_aspect	CEM analytic	15	0.400	0.058	0.533	0.000	0.962
long_bar_high_aspect	CEM analytic	15	0.333	0.053	0.667	0.000	0.752
long_bar_high_aspect	Robust perturb.	15	0.067	0.063	0.933	0.000	0.670
long_bar_high_aspect	Robust perturb.	15	0.000	0.060	1.000	0.000	0.552
long_bar_high_aspect	Robust perturb.	15	0.000	0.073	1.000	0.000	0.572
long_bar_high_aspect	Robust perturb.	15	0.000	0.068	1.000	0.000	0.569
long_bar_high_aspect	Robust perturb.	15	0.000	0.069	1.000	0.000	0.565
long_bar_high_aspect	Robust perturb.	15	0.067	0.063	0.933	0.000	0.638
long_bar_high_aspect	Robust perturb.	15	0.133	0.068	0.867	0.000	0.755
long_bar_high_aspect	Robust perturb.	15	0.000	0.062	1.000	0.000	0.565
long_bar_high_aspect	EBM v5	15	0.933	0.050	0.000	0.000	1.689
long_bar_high_aspect	EBM v5	15	0.800	0.071	0.067	0.000	1.731
long_bar_high_aspect	EBM v5	15	0.800	0.058	0.133	0.000	1.606
long_bar_high_aspect	EBM v5	15	0.733	0.062	0.133	0.000	1.549
long_bar_high_aspect	EBM v5	15	0.933	0.055	0.000	0.000	1.794
long_bar_high_aspect	EBM v5	15	0.867	0.053	0.067	0.000	1.733
long_bar_high_aspect	EBM v5	15	1.000	0.046	0.000	0.000	1.819
long_bar_high_aspect	EBM v5	15	1.000	0.044	0.000	0.000	1.616
long_bar_high_aspect	Oracle grid	15	0.933	0.035	0.067	0.000	1.155
long_bar_high_aspect	Oracle grid	15	0.800	0.051	0.067	0.000	1.280
long_bar_high_aspect	Oracle grid	15	0.867	0.033	0.133	0.000	1.307
long_bar_high_aspect	Oracle grid	15	0.867	0.041	0.067	0.000	1.791
long_bar_high_aspect	Oracle grid	15	0.800	0.056	0.067	0.000	1.638
long_bar_high_aspect	Oracle grid	15	0.933	0.044	0.067	0.000	1.323
long_bar_high_aspect	Oracle grid	15	1.000	0.033	0.000	0.000	1.634
long_bar_high_aspect	Oracle grid	15	1.000	0.037	0.000	0.000	1.400
material_shift_proxy	CEM analytic	15	0.200	0.093	0.800	0.000	1.824
material_shift_proxy	CEM analytic	15	0.133	0.088	0.600	0.000	2.009
material_shift_proxy	CEM analytic	15	0.333	0.073	0.600	0.000	2.273
material_shift_proxy	CEM analytic	15	0.467	0.073	0.400	0.000	2.172
material_shift_proxy	CEM analytic	15	0.133	0.081	0.800	0.000	1.902
material_shift_proxy	CEM analytic	15	0.200	0.071	0.800	0.000	1.942
material_shift_proxy	CEM analytic	15	0.400	0.075	0.600	0.000	2.065
material_shift_proxy	CEM analytic	15	0.133	0.092	0.867	0.000	1.901
material_shift_proxy	Robust perturb.	15	0.133	0.081	0.667	0.000	2.134

Split	Method	Episodes	Success	Slip	Drop	Collision	Energy
material_shift_proxy	Robust perturb.	15	0.200	0.087	0.733	0.000	1.962
material_shift_proxy	Robust perturb.	15	0.267	0.085	0.733	0.000	2.110
material_shift_proxy	Robust perturb.	15	0.267	0.079	0.733	0.000	2.008
material_shift_proxy	Robust perturb.	15	0.400	0.056	0.600	0.000	2.252
material_shift_proxy	Robust perturb.	15	0.133	0.080	0.867	0.000	1.845
material_shift_proxy	Robust perturb.	15	0.267	0.084	0.733	0.000	1.938
material_shift_proxy	Robust perturb.	15	0.267	0.071	0.733	0.000	1.982
material_shift_proxy	EBM v5	15	0.400	0.071	0.600	0.000	2.334
material_shift_proxy	EBM v5	15	0.800	0.050	0.133	0.000	2.610
material_shift_proxy	EBM v5	15	0.600	0.066	0.400	0.000	2.324
material_shift_proxy	EBM v5	15	0.600	0.075	0.333	0.000	2.373
material_shift_proxy	EBM v5	15	0.267	0.064	0.733	0.000	2.137
material_shift_proxy	EBM v5	15	0.267	0.070	0.733	0.000	2.102
material_shift_proxy	EBM v5	15	0.400	0.070	0.600	0.000	2.192
material_shift_proxy	EBM v5	15	0.600	0.058	0.400	0.000	2.420
material_shift_proxy	Oracle grid	15	0.533	0.068	0.267	0.000	2.465
material_shift_proxy	Oracle grid	15	0.667	0.052	0.267	0.000	2.517
material_shift_proxy	Oracle grid	15	0.667	0.059	0.200	0.000	2.528
material_shift_proxy	Oracle grid	15	0.733	0.059	0.133	0.000	2.506
material_shift_proxy	Oracle grid	15	0.733	0.048	0.200	0.000	2.693
material_shift_proxy	Oracle grid	15	0.667	0.053	0.333	0.000	2.219
material_shift_proxy	Oracle grid	15	0.800	0.045	0.200	0.000	2.429
material_shift_proxy	Oracle grid	15	0.667	0.055	0.333	0.000	2.482
seen_simple	CEM analytic	15	0.933	0.061	0.000	0.000	2.572
seen_simple	CEM analytic	15	0.867	0.056	0.000	0.000	2.711
seen_simple	CEM analytic	15	0.867	0.061	0.000	0.000	2.737
seen_simple	CEM analytic	15	0.800	0.061	0.000	0.000	2.693
seen_simple	CEM analytic	15	0.933	0.062	0.000	0.000	2.422
seen_simple	CEM analytic	15	0.867	0.051	0.000	0.000	2.614
seen_simple	CEM analytic	15	0.800	0.049	0.000	0.000	2.322
seen_simple	CEM analytic	15	0.933	0.045	0.000	0.000	2.532
seen_simple	Robust perturb.	15	1.000	0.056	0.000	0.000	2.747
seen_simple	Robust perturb.	15	0.867	0.068	0.000	0.000	2.730
seen_simple	Robust perturb.	15	1.000	0.043	0.000	0.000	2.769
seen_simple	Robust perturb.	15	0.800	0.066	0.000	0.000	3.046
seen_simple	Robust perturb.	15	1.000	0.064	0.000	0.000	2.752
seen_simple	Robust perturb.	15	0.933	0.051	0.000	0.000	2.673
seen_simple	Robust perturb.	15	0.933	0.047	0.000	0.000	2.859
seen_simple	Robust perturb.	15	0.933	0.050	0.000	0.000	2.590
seen_simple	EBM v5	15	1.000	0.039	0.000	0.000	2.844
seen_simple	EBM v5	15	1.000	0.043	0.000	0.000	2.692
seen_simple	EBM v5	15	1.000	0.036	0.000	0.000	2.569
seen_simple	EBM v5	15	1.000	0.037	0.000	0.000	2.790
seen_simple	EBM v5	15	1.000	0.044	0.000	0.000	2.800
seen_simple	EBM v5	15	1.000	0.046	0.000	0.000	2.816
seen_simple	EBM v5	15	1.000	0.042	0.000	0.000	2.720
seen_simple	EBM v5	15	1.000	0.036	0.000	0.000	2.794
seen_simple	Oracle grid	15	1.000	0.012	0.000	0.000	2.589
seen_simple	Oracle grid	15	1.000	0.017	0.000	0.000	2.516
seen_simple	Oracle grid	15	1.000	0.018	0.000	0.000	2.832
seen_simple	Oracle grid	15	1.000	0.014	0.000	0.000	2.588
seen_simple	Oracle grid	15	1.000	0.017	0.000	0.000	2.369
seen_simple	Oracle grid	15	1.000	0.012	0.000	0.000	2.605
seen_simple	Oracle grid	15	1.000	0.015	0.000	0.000	2.443
seen_simple	Oracle grid	15	1.000	0.018	0.000	0.000	2.579
slippery_contact	CEM analytic	15	0.333	0.082	0.600	0.000	2.129
slippery_contact	CEM analytic	15	0.400	0.070	0.600	0.000	2.005
slippery_contact	CEM analytic	15	0.133	0.096	0.867	0.000	1.828
slippery_contact	CEM analytic	15	0.467	0.084	0.533	0.000	1.999
slippery_contact	CEM analytic	15	0.067	0.098	0.800	0.000	1.943
slippery_contact	CEM analytic	15	0.400	0.071	0.467	0.000	2.187
slippery_contact	CEM analytic	15	0.333	0.074	0.667	0.000	1.989
slippery_contact	CEM analytic	15	0.333	0.082	0.533	0.000	2.138
slippery_contact	Robust perturb.	15	0.400	0.090	0.600	0.000	2.142
slippery_contact	Robust perturb.	15	0.267	0.084	0.600	0.000	2.107
slippery_contact	Robust perturb.	15	0.133	0.095	0.867	0.000	2.009
slippery_contact	Robust perturb.	15	0.200	0.075	0.800	0.000	1.766
slippery_contact	Robust perturb.	15	0.333	0.088	0.667	0.000	2.106
slippery_contact	Robust perturb.	15	0.400	0.075	0.533	0.000	2.079
slippery_contact	Robust perturb.	15	0.267	0.088	0.733	0.000	2.016
slippery_contact	Robust perturb.	15	0.467	0.069	0.467	0.000	2.238
slippery_contact	EBM v5	15	0.667	0.060	0.333	0.000	2.277
slippery_contact	EBM v5	15	0.733	0.068	0.267	0.000	2.288
slippery_contact	EBM v5	15	0.533	0.059	0.467	0.000	2.021
slippery_contact	EBM v5	15	0.533	0.067	0.467	0.000	2.194
slippery_contact	EBM v5	15	0.667	0.067	0.267	0.000	2.313
slippery_contact	EBM v5	15	0.400	0.082	0.467	0.000	2.045
slippery_contact	EBM v5	15	0.533	0.069	0.467	0.000	2.059
slippery_contact	EBM v5	15	0.600	0.055	0.400	0.000	2.156

Split	Method	Episodes	Success	Slip	Drop	Collision	Energy
slippery_contact	Oracle grid	15	0.733	0.059	0.200	0.000	2.507
slippery_contact	Oracle grid	15	0.933	0.040	0.067	0.000	2.553
slippery_contact	Oracle grid	15	0.733	0.061	0.200	0.000	2.478
slippery_contact	Oracle grid	15	0.867	0.053	0.133	0.000	2.317
slippery_contact	Oracle grid	15	0.867	0.056	0.133	0.000	2.437
slippery_contact	Oracle grid	15	0.933	0.046	0.067	0.000	2.647
slippery_contact	Oracle grid	15	0.800	0.066	0.067	0.000	2.558
slippery_contact	Oracle grid	15	0.933	0.045	0.067	0.000	2.567
task_constraint_shift	CEM analytic	15	0.200	0.069	0.800	0.000	0.729
task_constraint_shift	CEM analytic	15	0.333	0.065	0.600	0.000	1.017
task_constraint_shift	CEM analytic	15	0.333	0.059	0.667	0.000	0.843
task_constraint_shift	CEM analytic	15	0.267	0.060	0.667	0.000	0.912
task_constraint_shift	CEM analytic	15	0.200	0.061	0.800	0.000	0.767
task_constraint_shift	CEM analytic	15	0.333	0.062	0.667	0.000	0.881
task_constraint_shift	CEM analytic	15	0.400	0.068	0.467	0.000	1.352
task_constraint_shift	CEM analytic	15	0.400	0.055	0.600	0.000	0.877
task_constraint_shift	Robust perturb.	15	0.067	0.069	0.933	0.000	0.703
task_constraint_shift	Robust perturb.	15	0.200	0.071	0.800	0.000	0.882
task_constraint_shift	Robust perturb.	15	0.000	0.069	1.000	0.000	0.570
task_constraint_shift	Robust perturb.	15	0.000	0.065	1.000	0.000	0.567
task_constraint_shift	Robust perturb.	15	0.067	0.075	0.933	0.000	0.586
task_constraint_shift	Robust perturb.	15	0.200	0.059	0.800	0.000	0.792
task_constraint_shift	Robust perturb.	15	0.267	0.072	0.733	0.000	1.232
task_constraint_shift	Robust perturb.	15	0.067	0.066	0.933	0.000	0.596
task_constraint_shift	EBM v5	15	1.000	0.060	0.000	0.000	1.822
task_constraint_shift	EBM v5	15	1.000	0.052	0.000	0.000	1.826
task_constraint_shift	EBM v5	15	1.000	0.045	0.000	0.000	1.527
task_constraint_shift	EBM v5	15	0.933	0.051	0.000	0.000	1.728
task_constraint_shift	EBM v5	15	1.000	0.048	0.000	0.000	1.788
task_constraint_shift	EBM v5	15	1.000	0.039	0.000	0.000	1.785
task_constraint_shift	EBM v5	15	1.000	0.054	0.000	0.000	1.962
task_constraint_shift	EBM v5	15	1.000	0.039	0.000	0.000	1.573
task_constraint_shift	Oracle grid	15	0.800	0.053	0.133	0.000	1.729
task_constraint_shift	Oracle grid	15	0.933	0.048	0.000	0.000	1.733
task_constraint_shift	Oracle grid	15	0.867	0.041	0.067	0.000	2.084
task_constraint_shift	Oracle grid	15	0.933	0.034	0.067	0.000	1.133
task_constraint_shift	Oracle grid	15	0.933	0.041	0.067	0.000	1.291
task_constraint_shift	Oracle grid	15	0.933	0.032	0.067	0.000	1.490
task_constraint_shift	Oracle grid	15	0.933	0.031	0.067	0.000	1.774
task_constraint_shift	Oracle grid	15	0.867	0.037	0.067	0.000	1.713
thin_handle_shift	CEM analytic	15	0.933	0.051	0.000	0.000	2.193
thin_handle_shift	CEM analytic	15	0.867	0.049	0.000	0.000	2.050
thin_handle_shift	CEM analytic	15	0.933	0.056	0.000	0.000	2.251
thin_handle_shift	CEM analytic	15	0.733	0.062	0.000	0.000	2.092
thin_handle_shift	CEM analytic	15	1.000	0.056	0.000	0.000	2.154
thin_handle_shift	CEM analytic	15	0.733	0.064	0.000	0.000	2.248
thin_handle_shift	CEM analytic	15	0.933	0.054	0.000	0.000	2.114
thin_handle_shift	CEM analytic	15	0.933	0.052	0.000	0.000	2.236
thin_handle_shift	Robust perturb.	15	0.867	0.065	0.000	0.000	2.182
thin_handle_shift	Robust perturb.	15	0.933	0.039	0.000	0.000	2.286
thin_handle_shift	Robust perturb.	15	0.933	0.062	0.000	0.000	2.244
thin_handle_shift	Robust perturb.	15	0.733	0.066	0.000	0.000	2.052
thin_handle_shift	Robust perturb.	15	0.867	0.066	0.000	0.000	2.121
thin_handle_shift	Robust perturb.	15	0.733	0.061	0.000	0.000	2.246
thin_handle_shift	Robust perturb.	15	0.933	0.057	0.000	0.000	2.206
thin_handle_shift	Robust perturb.	15	0.933	0.036	0.000	0.000	2.241
thin_handle_shift	EBM v5	15	1.000	0.054	0.000	0.000	2.234
thin_handle_shift	EBM v5	15	1.000	0.036	0.000	0.000	2.227
thin_handle_shift	EBM v5	15	1.000	0.056	0.000	0.000	2.222
thin_handle_shift	EBM v5	15	1.000	0.041	0.000	0.000	2.077
thin_handle_shift	EBM v5	15	1.000	0.055	0.000	0.000	2.161
thin_handle_shift	EBM v5	15	0.867	0.052	0.000	0.000	2.212
thin_handle_shift	EBM v5	15	1.000	0.049	0.000	0.000	2.260
thin_handle_shift	EBM v5	15	0.933	0.036	0.000	0.000	2.237
thin_handle_shift	Oracle grid	15	1.000	0.018	0.000	0.000	1.623
thin_handle_shift	Oracle grid	15	1.000	0.014	0.000	0.000	1.667
thin_handle_shift	Oracle grid	15	1.000	0.023	0.000	0.000	1.670
thin_handle_shift	Oracle grid	15	1.000	0.024	0.000	0.000	1.522
thin_handle_shift	Oracle grid	15	1.000	0.019	0.000	0.000	1.546
thin_handle_shift	Oracle grid	15	1.000	0.023	0.000	0.000	1.621
thin_handle_shift	Oracle grid	15	1.000	0.021	0.000	0.000	1.482
thin_handle_shift	Oracle grid	15	1.000	0.022	0.000	0.000	1.883
unseen_dimensions	CEM analytic	15	0.600	0.067	0.333	0.000	3.270
unseen_dimensions	CEM analytic	15	0.733	0.049	0.067	0.000	4.214
unseen_dimensions	CEM analytic	15	0.733	0.059	0.133	0.000	4.201
unseen_dimensions	CEM analytic	15	0.800	0.061	0.000	0.000	4.597
unseen_dimensions	CEM analytic	15	0.533	0.060	0.333	0.000	3.369
unseen_dimensions	CEM analytic	15	0.667	0.069	0.200	0.000	3.650
unseen_dimensions	CEM analytic	15	0.600	0.078	0.200	0.000	4.049

Split	Method	Episodes	Success	Slip	Drop	Collision	Energy
unseen_dimensions	CEM analytic	15	0.467	0.070	0.400	0.000	2.968
unseen_dimensions	Robust perturb.	15	0.733	0.064	0.267	0.000	3.562
unseen_dimensions	Robust perturb.	15	0.800	0.050	0.067	0.000	4.051
unseen_dimensions	Robust perturb.	15	0.800	0.063	0.133	0.000	4.150
unseen_dimensions	Robust perturb.	15	0.733	0.066	0.267	0.000	3.779
unseen_dimensions	Robust perturb.	15	0.667	0.064	0.200	0.000	3.846
unseen_dimensions	Robust perturb.	15	0.800	0.064	0.067	0.000	3.714
unseen_dimensions	Robust perturb.	15	0.600	0.077	0.333	0.000	3.572
unseen_dimensions	Robust perturb.	15	0.467	0.070	0.533	0.000	2.632
unseen_dimensions	EBM v5	15	1.000	0.034	0.000	0.000	2.601
unseen_dimensions	EBM v5	15	1.000	0.033	0.000	0.000	3.323
unseen_dimensions	EBM v5	15	1.000	0.037	0.000	0.000	4.036
unseen_dimensions	EBM v5	15	0.933	0.032	0.067	0.000	2.900
unseen_dimensions	EBM v5	15	0.867	0.042	0.000	0.000	3.133
unseen_dimensions	EBM v5	15	1.000	0.043	0.000	0.000	3.091
unseen_dimensions	EBM v5	15	1.000	0.041	0.000	0.000	3.038
unseen_dimensions	EBM v5	15	0.933	0.051	0.067	0.000	2.626
unseen_dimensions	Oracle grid	15	1.000	0.026	0.000	0.000	3.516
unseen_dimensions	Oracle grid	15	1.000	0.016	0.000	0.000	3.059
unseen_dimensions	Oracle grid	15	1.000	0.024	0.000	0.000	2.638
unseen_dimensions	Oracle grid	15	0.933	0.025	0.000	0.000	2.771
unseen_dimensions	Oracle grid	15	1.000	0.030	0.000	0.000	3.959
unseen_dimensions	Oracle grid	15	1.000	0.026	0.000	0.000	2.707
unseen_dimensions	Oracle grid	15	1.000	0.028	0.000	0.000	2.844
unseen_dimensions	Oracle grid	15	0.933	0.026	0.067	0.000	3.332
unseen_shape_family	CEM analytic	15	1.000	0.058	0.000	0.000	3.718
unseen_shape_family	CEM analytic	15	0.933	0.058	0.000	0.000	3.524
unseen_shape_family	CEM analytic	15	0.933	0.060	0.000	0.000	3.567
unseen_shape_family	CEM analytic	15	0.867	0.061	0.000	0.000	3.534
unseen_shape_family	CEM analytic	15	1.000	0.050	0.000	0.000	3.754
unseen_shape_family	CEM analytic	15	0.733	0.072	0.000	0.000	3.789
unseen_shape_family	CEM analytic	15	1.000	0.065	0.000	0.000	3.494
unseen_shape_family	CEM analytic	15	0.933	0.046	0.000	0.000	3.507
unseen_shape_family	Robust perturb.	15	1.000	0.066	0.000	0.000	3.885
unseen_shape_family	Robust perturb.	15	0.800	0.069	0.000	0.000	3.638
unseen_shape_family	Robust perturb.	15	0.933	0.056	0.000	0.000	3.604
unseen_shape_family	Robust perturb.	15	0.933	0.055	0.000	0.000	3.689
unseen_shape_family	Robust perturb.	15	1.000	0.053	0.000	0.000	3.834
unseen_shape_family	Robust perturb.	15	0.800	0.065	0.000	0.000	3.840
unseen_shape_family	Robust perturb.	15	1.000	0.064	0.000	0.000	3.675
unseen_shape_family	Robust perturb.	15	0.933	0.052	0.000	0.000	3.539
unseen_shape_family	EBM v5	15	1.000	0.058	0.000	0.000	3.709
unseen_shape_family	EBM v5	15	0.933	0.052	0.000	0.000	3.572
unseen_shape_family	EBM v5	15	1.000	0.045	0.000	0.000	3.598
unseen_shape_family	EBM v5	15	0.933	0.050	0.000	0.000	3.645
unseen_shape_family	EBM v5	15	1.000	0.056	0.000	0.000	3.715
unseen_shape_family	EBM v5	15	0.867	0.060	0.000	0.000	3.822
unseen_shape_family	EBM v5	15	1.000	0.053	0.000	0.000	3.559
unseen_shape_family	EBM v5	15	1.000	0.035	0.000	0.000	3.459
unseen_shape_family	Oracle grid	15	1.000	0.020	0.000	0.000	2.580
unseen_shape_family	Oracle grid	15	1.000	0.020	0.000	0.000	2.817
unseen_shape_family	Oracle grid	15	1.000	0.015	0.000	0.000	2.762
unseen_shape_family	Oracle grid	15	1.000	0.016	0.000	0.000	3.116
unseen_shape_family	Oracle grid	15	1.000	0.013	0.000	0.000	2.862
unseen_shape_family	Oracle grid	15	1.000	0.016	0.000	0.000	2.795
unseen_shape_family	Oracle grid	15	1.000	0.017	0.000	0.000	2.885
unseen_shape_family	Oracle grid	15	1.000	0.013	0.000	0.000	3.091

F ALL SEED METRICS

Table 11: All method/split/seed metrics from the frozen run.

Split	Method	Episodes	Success	Slip	Drop	Collision	Energy
adversarial_clutter_gap	Random	15	0.000	0.083	0.733	1.000	2.092
adversarial_clutter_gap	Random	15	0.000	0.087	0.867	1.000	1.565
adversarial_clutter_gap	Random	15	0.000	0.083	0.867	1.000	1.870
adversarial_clutter_gap	Random	15	0.067	0.089	0.867	0.933	2.412
adversarial_clutter_gap	Random	15	0.000	0.091	0.867	1.000	1.771
adversarial_clutter_gap	Random	15	0.000	0.085	0.933	1.000	1.899
adversarial_clutter_gap	Random	15	0.000	0.094	0.800	1.000	1.805
adversarial_clutter_gap	Random	15	0.000	0.090	0.800	1.000	2.311
adversarial_clutter_gap	Antipodal	15	0.000	0.089	0.933	1.000	1.677
adversarial_clutter_gap	Antipodal	15	0.000	0.079	0.533	0.933	3.179
adversarial_clutter_gap	Antipodal	15	0.000	0.094	0.800	1.000	2.917

Split	Method	Episodes	Success	Slip	Drop	Collision	Energy
adversarial_clutter_gap	Antipodal	15	0.000	0.105	0.933	0.933	1.924
adversarial_clutter_gap	Antipodal	15	0.000	0.094	0.800	1.000	2.682
adversarial_clutter_gap	Antipodal	15	0.000	0.073	0.533	1.000	3.689
adversarial_clutter_gap	Antipodal	15	0.000	0.090	0.867	1.000	2.319
adversarial_clutter_gap	Antipodal	15	0.000	0.078	0.800	1.000	2.983
adversarial_clutter_gap	Force closure	15	0.000	0.071	0.867	1.000	2.087
adversarial_clutter_gap	Force closure	15	0.000	0.067	0.400	1.000	3.547
adversarial_clutter_gap	Force closure	15	0.000	0.081	0.733	1.000	3.204
adversarial_clutter_gap	Force closure	15	0.000	0.103	0.867	0.933	2.158
adversarial_clutter_gap	Force closure	15	0.000	0.082	0.800	1.000	2.750
adversarial_clutter_gap	Force closure	15	0.000	0.071	0.600	1.000	3.657
adversarial_clutter_gap	Force closure	15	0.000	0.082	0.800	1.000	2.591
adversarial_clutter_gap	Force closure	15	0.000	0.081	0.800	1.000	2.801
adversarial_clutter_gap	Collision-aware FC	15	0.000	0.073	0.733	0.867	3.217
adversarial_clutter_gap	Collision-aware FC	15	0.000	0.066	0.400	1.000	3.439
adversarial_clutter_gap	Collision-aware FC	15	0.000	0.090	0.733	1.000	3.070
adversarial_clutter_gap	Collision-aware FC	15	0.000	0.097	0.867	1.000	3.022
adversarial_clutter_gap	Collision-aware FC	15	0.000	0.072	0.733	1.000	2.899
adversarial_clutter_gap	Collision-aware FC	15	0.067	0.101	0.667	0.933	3.898
adversarial_clutter_gap	Collision-aware FC	15	0.000	0.087	0.733	1.000	2.583
adversarial_clutter_gap	Collision-aware FC	15	0.000	0.070	0.867	1.000	3.129
adversarial_clutter_gap	CEM analytic	15	0.000	0.073	0.800	0.867	2.671
adversarial_clutter_gap	CEM analytic	15	0.000	0.064	0.400	1.000	3.529
adversarial_clutter_gap	CEM analytic	15	0.000	0.086	0.800	1.000	3.179
adversarial_clutter_gap	CEM analytic	15	0.000	0.111	0.800	0.933	2.575
adversarial_clutter_gap	CEM analytic	15	0.000	0.085	0.733	1.000	2.905
adversarial_clutter_gap	CEM analytic	15	0.000	0.100	0.533	1.000	4.426
adversarial_clutter_gap	CEM analytic	15	0.000	0.085	0.733	1.000	2.553
adversarial_clutter_gap	CEM analytic	15	0.000	0.064	0.733	1.000	3.406
adversarial_clutter_gap	Robust perturb.	15	0.000	0.075	0.733	0.933	3.204
adversarial_clutter_gap	Robust perturb.	15	0.000	0.076	0.467	1.000	3.337
adversarial_clutter_gap	Robust perturb.	15	0.000	0.090	0.800	1.000	2.720
adversarial_clutter_gap	Robust perturb.	15	0.000	0.094	0.867	1.000	2.987
adversarial_clutter_gap	Robust perturb.	15	0.000	0.083	0.733	1.000	2.800
adversarial_clutter_gap	Robust perturb.	15	0.000	0.103	0.733	1.000	3.600
adversarial_clutter_gap	Robust perturb.	15	0.000	0.093	0.800	1.000	2.411
adversarial_clutter_gap	Robust perturb.	15	0.000	0.075	0.733	1.000	3.372
adversarial_clutter_gap	MLP energy	15	0.000	0.090	0.867	1.000	2.313
adversarial_clutter_gap	MLP energy	15	0.000	0.092	0.800	1.000	2.682
adversarial_clutter_gap	MLP energy	15	0.000	0.086	0.800	1.000	2.336
adversarial_clutter_gap	MLP energy	15	0.000	0.107	0.800	1.000	2.198
adversarial_clutter_gap	MLP energy	15	0.000	0.082	0.800	1.000	2.228
adversarial_clutter_gap	MLP energy	15	0.000	0.081	0.867	1.000	2.055
adversarial_clutter_gap	MLP energy	15	0.000	0.082	0.800	1.000	2.683
adversarial_clutter_gap	MLP energy	15	0.000	0.086	0.800	1.000	2.583
adversarial_clutter_gap	HGB energy	15	0.000	0.078	0.933	1.000	2.181
adversarial_clutter_gap	HGB energy	15	0.000	0.092	0.467	0.933	3.359
adversarial_clutter_gap	HGB energy	15	0.000	0.093	0.733	1.000	2.455
adversarial_clutter_gap	HGB energy	15	0.000	0.101	0.867	0.933	1.825
adversarial_clutter_gap	HGB energy	15	0.000	0.070	0.733	1.000	2.870
adversarial_clutter_gap	HGB energy	15	0.000	0.086	0.800	0.933	2.051
adversarial_clutter_gap	HGB energy	15	0.000	0.086	0.667	1.000	2.103
adversarial_clutter_gap	HGB energy	15	0.000	0.103	0.800	1.000	2.608
adversarial_clutter_gap	Transformer	15	0.000	0.086	0.800	0.933	2.079
adversarial_clutter_gap	Transformer	15	0.000	0.088	0.800	1.000	2.093
adversarial_clutter_gap	Transformer	15	0.000	0.080	0.800	1.000	1.674
adversarial_clutter_gap	Transformer	15	0.000	0.113	0.733	1.000	1.965
adversarial_clutter_gap	Transformer	15	0.000	0.072	0.867	1.000	1.883
adversarial_clutter_gap	Transformer	15	0.000	0.065	0.733	1.000	2.256
adversarial_clutter_gap	Transformer	15	0.000	0.093	0.800	1.000	2.382
adversarial_clutter_gap	Transformer	15	0.000	0.085	0.867	1.000	2.236
adversarial_clutter_gap	Ensemble	15	0.000	0.087	0.800	0.933	2.141
adversarial_clutter_gap	Ensemble	15	0.000	0.094	0.600	1.000	3.493
adversarial_clutter_gap	Ensemble	15	0.000	0.083	0.533	1.000	3.235
adversarial_clutter_gap	Ensemble	15	0.000	0.098	0.867	1.000	2.748
adversarial_clutter_gap	Ensemble	15	0.000	0.075	0.733	1.000	2.938
adversarial_clutter_gap	Ensemble	15	0.000	0.108	0.933	1.000	2.550
adversarial_clutter_gap	Ensemble	15	0.000	0.114	0.933	1.000	1.790
adversarial_clutter_gap	Ensemble	15	0.000	0.105	0.800	1.000	2.329
adversarial_clutter_gap	Calibrated stack	15	0.000	0.084	0.800	0.933	3.007
adversarial_clutter_gap	Calibrated stack	15	0.000	0.084	0.667	1.000	2.861
adversarial_clutter_gap	Calibrated stack	15	0.000	0.084	0.600	1.000	2.987
adversarial_clutter_gap	Calibrated stack	15	0.000	0.113	0.867	1.000	2.777
adversarial_clutter_gap	Calibrated stack	15	0.000	0.084	0.733	1.000	2.843
adversarial_clutter_gap	Calibrated stack	15	0.000	0.104	0.667	1.000	3.724
adversarial_clutter_gap	Calibrated stack	15	0.000	0.089	0.800	1.000	2.474
adversarial_clutter_gap	Calibrated stack	15	0.000	0.076	0.867	1.000	2.731
adversarial_clutter_gap	Old EBM	15	0.000	0.074	0.800	0.933	2.556
adversarial_clutter_gap	Old EBM	15	0.000	0.075	0.467	1.000	3.463

Split	Method	Episodes	Success	Slip	Drop	Collision	Energy
adversarial_clutter_gap	Old EBM	15	0.000	0.084	0.667	1.000	3.391
adversarial_clutter_gap	Old EBM	15	0.000	0.103	0.800	1.000	2.627
adversarial_clutter_gap	Old EBM	15	0.067	0.077	0.733	0.933	3.150
adversarial_clutter_gap	Old EBM	15	0.067	0.101	0.600	0.933	3.862
adversarial_clutter_gap	Old EBM	15	0.000	0.097	0.867	1.000	2.206
adversarial_clutter_gap	Old EBM	15	0.000	0.074	0.733	1.000	3.109
adversarial_clutter_gap	EBM v5	15	0.000	0.081	0.800	0.933	3.130
adversarial_clutter_gap	EBM v5	15	0.000	0.079	0.533	1.000	3.253
adversarial_clutter_gap	EBM v5	15	0.000	0.091	0.733	1.000	2.921
adversarial_clutter_gap	EBM v5	15	0.000	0.094	0.867	1.000	2.977
adversarial_clutter_gap	EBM v5	15	0.000	0.082	0.733	1.000	2.848
adversarial_clutter_gap	EBM v5	15	0.067	0.105	0.667	0.933	3.801
adversarial_clutter_gap	EBM v5	15	0.000	0.092	0.800	1.000	2.458
adversarial_clutter_gap	EBM v5	15	0.000	0.077	0.800	1.000	3.095
adversarial_clutter_gap	Oracle grid	15	0.000	0.060	0.333	0.933	3.897
adversarial_clutter_gap	Oracle grid	15	0.000	0.047	0.067	1.000	4.209
adversarial_clutter_gap	Oracle grid	15	0.000	0.045	0.133	1.000	5.164
adversarial_clutter_gap	Oracle grid	15	0.133	0.071	0.200	0.800	4.358
adversarial_clutter_gap	Oracle grid	15	0.133	0.034	0.200	0.867	4.571
adversarial_clutter_gap	Oracle grid	15	0.133	0.039	0.133	0.867	4.439
adversarial_clutter_gap	Oracle grid	15	0.000	0.053	0.333	1.000	3.594
adversarial_clutter_gap	Oracle grid	15	0.133	0.057	0.467	0.733	3.837
clutter_collision	Random	15	0.067	0.059	0.467	0.933	1.911
clutter_collision	Random	15	0.200	0.068	0.400	0.800	1.694
clutter_collision	Random	15	0.067	0.075	0.600	0.933	1.615
clutter_collision	Random	15	0.000	0.067	0.533	1.000	1.593
clutter_collision	Random	15	0.133	0.061	0.400	0.867	1.728
clutter_collision	Random	15	0.133	0.068	0.467	0.867	1.903
clutter_collision	Random	15	0.067	0.059	0.400	0.933	1.577
clutter_collision	Random	15	0.067	0.060	0.400	0.933	2.155
clutter_collision	Antipodal	15	0.400	0.075	0.133	0.533	2.476
clutter_collision	Antipodal	15	0.267	0.061	0.067	0.733	2.943
clutter_collision	Antipodal	15	0.400	0.054	0.067	0.600	2.804
clutter_collision	Antipodal	15	0.133	0.051	0.267	0.867	2.204
clutter_collision	Antipodal	15	0.333	0.058	0.200	0.600	2.590
clutter_collision	Antipodal	15	0.333	0.055	0.067	0.533	2.939
clutter_collision	Antipodal	15	0.467	0.047	0.200	0.533	2.688
clutter_collision	Antipodal	15	0.333	0.059	0.133	0.667	2.629
clutter_collision	Force closure	15	0.467	0.059	0.200	0.533	2.586
clutter_collision	Force closure	15	0.200	0.066	0.200	0.800	2.552
clutter_collision	Force closure	15	0.333	0.058	0.200	0.667	2.639
clutter_collision	Force closure	15	0.133	0.050	0.267	0.800	2.383
clutter_collision	Force closure	15	0.267	0.054	0.200	0.733	2.715
clutter_collision	Force closure	15	0.333	0.058	0.133	0.600	2.773
clutter_collision	Force closure	15	0.533	0.045	0.133	0.467	2.813
clutter_collision	Force closure	15	0.400	0.058	0.333	0.600	2.630
clutter_collision	Collision-aware FC	15	0.267	0.081	0.200	0.600	2.524
clutter_collision	Collision-aware FC	15	0.267	0.064	0.133	0.733	2.633
clutter_collision	Collision-aware FC	15	0.400	0.058	0.067	0.600	2.700
clutter_collision	Collision-aware FC	15	0.200	0.064	0.400	0.733	2.103
clutter_collision	Collision-aware FC	15	0.333	0.055	0.067	0.667	2.904
clutter_collision	Collision-aware FC	15	0.467	0.059	0.133	0.400	2.855
clutter_collision	Collision-aware FC	15	0.533	0.045	0.133	0.467	2.813
clutter_collision	Collision-aware FC	15	0.400	0.058	0.133	0.600	2.691
clutter_collision	CEM analytic	15	0.333	0.083	0.133	0.533	2.565
clutter_collision	CEM analytic	15	0.267	0.059	0.133	0.733	2.755
clutter_collision	CEM analytic	15	0.400	0.058	0.067	0.600	2.709
clutter_collision	CEM analytic	15	0.200	0.064	0.400	0.733	2.107
clutter_collision	CEM analytic	15	0.200	0.053	0.067	0.800	2.864
clutter_collision	CEM analytic	15	0.467	0.058	0.133	0.400	2.850
clutter_collision	CEM analytic	15	0.600	0.039	0.133	0.400	2.801
clutter_collision	CEM analytic	15	0.400	0.057	0.133	0.600	2.826
clutter_collision	Robust perturb.	15	0.200	0.081	0.200	0.667	2.510
clutter_collision	Robust perturb.	15	0.267	0.069	0.133	0.733	2.673
clutter_collision	Robust perturb.	15	0.333	0.058	0.067	0.667	2.478
clutter_collision	Robust perturb.	15	0.200	0.064	0.333	0.733	2.235
clutter_collision	Robust perturb.	15	0.333	0.055	0.067	0.667	2.902
clutter_collision	Robust perturb.	15	0.533	0.055	0.133	0.333	2.850
clutter_collision	Robust perturb.	15	0.533	0.049	0.133	0.467	2.865
clutter_collision	Robust perturb.	15	0.400	0.062	0.200	0.600	2.704
clutter_collision	MLP energy	15	0.600	0.056	0.200	0.400	2.297
clutter_collision	MLP energy	15	0.467	0.034	0.133	0.533	2.353
clutter_collision	MLP energy	15	0.333	0.056	0.133	0.667	2.352
clutter_collision	MLP energy	15	0.400	0.050	0.267	0.533	2.019
clutter_collision	MLP energy	15	0.533	0.053	0.133	0.467	2.247
clutter_collision	MLP energy	15	0.533	0.039	0.067	0.467	2.429
clutter_collision	MLP energy	15	0.600	0.031	0.067	0.400	2.657
clutter_collision	MLP energy	15	0.467	0.045	0.267	0.533	2.159
clutter_collision	HGB energy	15	0.467	0.052	0.133	0.533	2.368

Split	Method	Episodes	Success	Slip	Drop	Collision	Energy
clutter_collision	HGB energy	15	0.533	0.048	0.067	0.400	2.503
clutter_collision	HGB energy	15	0.333	0.049	0.133	0.667	2.402
clutter_collision	HGB energy	15	0.467	0.056	0.133	0.533	2.204
clutter_collision	HGB energy	15	0.533	0.052	0.133	0.400	2.284
clutter_collision	HGB energy	15	0.600	0.038	0.000	0.400	2.718
clutter_collision	HGB energy	15	0.333	0.043	0.267	0.667	2.449
clutter_collision	HGB energy	15	0.267	0.053	0.200	0.733	2.236
clutter_collision	Transformer	15	0.467	0.061	0.067	0.533	2.594
clutter_collision	Transformer	15	0.333	0.061	0.067	0.667	2.750
clutter_collision	Transformer	15	0.400	0.059	0.000	0.600	2.592
clutter_collision	Transformer	15	0.267	0.068	0.267	0.600	2.117
clutter_collision	Transformer	15	0.533	0.056	0.067	0.467	2.547
clutter_collision	Transformer	15	0.333	0.047	0.067	0.600	2.798
clutter_collision	Transformer	15	0.533	0.052	0.133	0.467	2.731
clutter_collision	Transformer	15	0.333	0.059	0.133	0.667	2.671
clutter_collision	Ensemble	15	0.400	0.056	0.267	0.533	2.084
clutter_collision	Ensemble	15	0.267	0.068	0.067	0.667	2.652
clutter_collision	Ensemble	15	0.133	0.073	0.200	0.800	2.167
clutter_collision	Ensemble	15	0.333	0.059	0.200	0.600	2.072
clutter_collision	Ensemble	15	0.600	0.042	0.000	0.400	2.466
clutter_collision	Ensemble	15	0.467	0.050	0.133	0.467	2.582
clutter_collision	Ensemble	15	0.600	0.050	0.067	0.400	2.750
clutter_collision	Ensemble	15	0.533	0.051	0.200	0.467	2.327
clutter_collision	Calibrated stack	15	0.467	0.060	0.200	0.467	2.082
clutter_collision	Calibrated stack	15	0.533	0.046	0.133	0.467	2.465
clutter_collision	Calibrated stack	15	0.333	0.049	0.067	0.667	2.533
clutter_collision	Calibrated stack	15	0.400	0.049	0.267	0.533	2.041
clutter_collision	Calibrated stack	15	0.467	0.056	0.067	0.533	2.581
clutter_collision	Calibrated stack	15	0.467	0.036	0.067	0.533	2.732
clutter_collision	Calibrated stack	15	0.600	0.036	0.067	0.400	2.813
clutter_collision	Calibrated stack	15	0.533	0.044	0.200	0.467	2.291
clutter_collision	Old EBM	15	0.333	0.073	0.067	0.533	2.641
clutter_collision	Old EBM	15	0.267	0.061	0.133	0.733	2.674
clutter_collision	Old EBM	15	0.333	0.057	0.000	0.667	2.652
clutter_collision	Old EBM	15	0.267	0.063	0.333	0.667	2.178
clutter_collision	Old EBM	15	0.467	0.057	0.067	0.533	2.668
clutter_collision	Old EBM	15	0.533	0.043	0.067	0.467	2.848
clutter_collision	Old EBM	15	0.533	0.043	0.133	0.467	2.760
clutter_collision	Old EBM	15	0.333	0.059	0.133	0.667	2.686
clutter_collision	EBM v5	15	0.467	0.071	0.133	0.467	2.329
clutter_collision	EBM v5	15	0.267	0.061	0.133	0.733	2.674
clutter_collision	EBM v5	15	0.333	0.051	0.067	0.667	2.566
clutter_collision	EBM v5	15	0.333	0.058	0.267	0.600	2.292
clutter_collision	EBM v5	15	0.467	0.057	0.067	0.533	2.668
clutter_collision	EBM v5	15	0.467	0.039	0.067	0.533	2.806
clutter_collision	EBM v5	15	0.533	0.040	0.133	0.467	2.814
clutter_collision	EBM v5	15	0.400	0.058	0.200	0.600	2.559
clutter_collision	Oracle grid	15	0.800	0.029	0.000	0.200	2.673
clutter_collision	Oracle grid	15	0.933	0.033	0.000	0.067	2.562
clutter_collision	Oracle grid	15	0.733	0.047	0.000	0.200	2.775
clutter_collision	Oracle grid	15	0.867	0.046	0.000	0.133	2.643
clutter_collision	Oracle grid	15	0.933	0.044	0.000	0.067	2.344
clutter_collision	Oracle grid	15	0.933	0.029	0.000	0.067	2.563
clutter_collision	Oracle grid	15	0.933	0.019	0.000	0.067	2.729
clutter_collision	Oracle grid	15	0.867	0.031	0.000	0.133	2.544
combined_composition_shift	Random	15	0.000	0.074	0.733	1.000	2.174
combined_composition_shift	Random	15	0.000	0.083	0.667	0.933	2.076
combined_composition_shift	Random	15	0.000	0.092	0.733	1.000	1.868
combined_composition_shift	Random	15	0.000	0.091	0.667	1.000	1.951
combined_composition_shift	Random	15	0.000	0.102	0.667	1.000	2.121
combined_composition_shift	Random	15	0.067	0.068	0.333	0.933	3.385
combined_composition_shift	Random	15	0.000	0.074	0.600	1.000	2.806
combined_composition_shift	Random	15	0.000	0.091	0.800	1.000	1.366
combined_composition_shift	Antipodal	15	0.000	0.129	0.667	0.867	2.741
combined_composition_shift	Antipodal	15	0.000	0.109	0.400	0.867	3.358
combined_composition_shift	Antipodal	15	0.067	0.079	0.200	0.867	4.840
combined_composition_shift	Antipodal	15	0.067	0.090	0.467	0.800	3.410
combined_composition_shift	Antipodal	15	0.000	0.111	0.533	0.933	3.256
combined_composition_shift	Antipodal	15	0.067	0.076	0.333	0.867	3.564
combined_composition_shift	Antipodal	15	0.133	0.064	0.133	0.867	4.584
combined_composition_shift	Antipodal	15	0.067	0.087	0.467	0.867	3.380
combined_composition_shift	Force closure	15	0.000	0.129	0.667	0.867	2.739
combined_composition_shift	Force closure	15	0.000	0.096	0.400	1.000	3.325
combined_composition_shift	Force closure	15	0.067	0.076	0.200	0.867	4.770
combined_composition_shift	Force closure	15	0.133	0.085	0.400	0.800	3.754
combined_composition_shift	Force closure	15	0.067	0.101	0.600	0.867	3.060
combined_composition_shift	Force closure	15	0.067	0.076	0.333	0.867	3.566
combined_composition_shift	Force closure	15	0.133	0.063	0.133	0.867	4.598
combined_composition_shift	Force closure	15	0.067	0.082	0.467	0.867	3.433

	Split	Method	Episodes	Success	Slip	Drop	Collision	Energy
1080								
1081								
1082	combined_composition_shift	Collision-aware FC	15	0.000	0.087	0.467	0.933	3.406
	combined_composition_shift	Collision-aware FC	15	0.000	0.099	0.400	1.000	3.549
1083	combined_composition_shift	Collision-aware FC	15	0.067	0.089	0.267	0.867	4.271
1084	combined_composition_shift	Collision-aware FC	15	0.133	0.081	0.333	0.800	3.829
	combined_composition_shift	Collision-aware FC	15	0.000	0.112	0.667	0.933	2.695
1085	combined_composition_shift	Collision-aware FC	15	0.067	0.084	0.467	0.867	3.453
1086	combined_composition_shift	Collision-aware FC	15	0.067	0.077	0.467	0.933	3.152
	combined_composition_shift	Collision-aware FC	15	0.067	0.083	0.333	0.867	3.395
1087	combined_composition_shift	CEM analytic	15	0.000	0.106	0.400	0.867	4.055
	combined_composition_shift	CEM analytic	15	0.000	0.105	0.333	1.000	3.650
1088	combined_composition_shift	CEM analytic	15	0.067	0.072	0.133	0.867	4.875
1089	combined_composition_shift	CEM analytic	15	0.067	0.089	0.467	0.867	3.389
	combined_composition_shift	CEM analytic	15	0.000	0.105	0.533	0.933	3.038
1090	combined_composition_shift	CEM analytic	15	0.067	0.078	0.333	0.867	3.754
1091	combined_composition_shift	CEM analytic	15	0.067	0.070	0.400	0.933	3.859
	combined_composition_shift	CEM analytic	15	0.067	0.088	0.467	0.867	3.130
1092	combined_composition_shift	Robust perturb.	15	0.000	0.085	0.533	0.933	3.309
1093	combined_composition_shift	Robust perturb.	15	0.000	0.104	0.400	0.933	3.535
	combined_composition_shift	Robust perturb.	15	0.067	0.089	0.267	0.867	4.270
1094	combined_composition_shift	Robust perturb.	15	0.067	0.087	0.400	0.800	3.232
1095	combined_composition_shift	Robust perturb.	15	0.000	0.106	0.667	0.867	2.701
	combined_composition_shift	Robust perturb.	15	0.067	0.084	0.467	0.867	3.453
1096	combined_composition_shift	Robust perturb.	15	0.067	0.075	0.533	0.933	2.921
1097	combined_composition_shift	Robust perturb.	15	0.067	0.075	0.200	0.933	3.718
	combined_composition_shift	MLP energy	15	0.000	0.070	0.600	1.000	2.611
1098	combined_composition_shift	MLP energy	15	0.000	0.073	0.400	1.000	2.590
1099	combined_composition_shift	MLP energy	15	0.067	0.087	0.533	0.933	2.527
	combined_composition_shift	MLP energy	15	0.067	0.057	0.467	0.933	2.853
1100	combined_composition_shift	MLP energy	15	0.067	0.098	0.667	0.933	2.546
1101	combined_composition_shift	MLP energy	15	0.067	0.068	0.333	0.933	3.530
	combined_composition_shift	MLP energy	15	0.067	0.061	0.533	0.933	2.803
1102	combined_composition_shift	MLP energy	15	0.067	0.071	0.333	0.933	3.351
1103	combined_composition_shift	HGB energy	15	0.000	0.107	0.600	0.867	3.292
	combined_composition_shift	HGB energy	15	0.000	0.119	0.600	0.933	2.546
1104	combined_composition_shift	HGB energy	15	0.200	0.071	0.333	0.667	4.035
1105	combined_composition_shift	HGB energy	15	0.067	0.075	0.400	0.867	2.989
	combined_composition_shift	HGB energy	15	0.000	0.107	0.733	1.000	2.210
1106	combined_composition_shift	HGB energy	15	0.000	0.093	0.667	0.867	2.788
1107	combined_composition_shift	HGB energy	15	0.133	0.057	0.200	0.867	4.272
	combined_composition_shift	HGB energy	15	0.133	0.067	0.400	0.867	3.649
1108	combined_composition_shift	Transformer	15	0.000	0.078	0.667	0.933	2.825
1109	combined_composition_shift	Transformer	15	0.000	0.084	0.667	0.933	2.170
	combined_composition_shift	Transformer	15	0.067	0.072	0.467	0.933	2.266
1110	combined_composition_shift	Transformer	15	0.000	0.084	0.733	0.867	1.905
1111	combined_composition_shift	Transformer	15	0.067	0.088	0.533	0.867	2.723
	combined_composition_shift	Transformer	15	0.000	0.061	0.333	1.000	3.057
1112	combined_composition_shift	Transformer	15	0.067	0.053	0.200	0.933	3.933
1113	combined_composition_shift	Transformer	15	0.000	0.069	0.467	1.000	3.041
	combined_composition_shift	Ensemble	15	0.000	0.091	0.533	0.933	3.523
1114	combined_composition_shift	Ensemble	15	0.000	0.087	0.467	0.933	3.181
1115	combined_composition_shift	Ensemble	15	0.067	0.102	0.533	0.867	3.138
	combined_composition_shift	Ensemble	15	0.067	0.087	0.533	0.800	3.121
1116	combined_composition_shift	Ensemble	15	0.067	0.116	0.733	0.800	2.500
1117	combined_composition_shift	Ensemble	15	0.067	0.074	0.533	0.867	3.373
	combined_composition_shift	Ensemble	15	0.200	0.053	0.200	0.800	4.431
1118	combined_composition_shift	Ensemble	15	0.133	0.077	0.467	0.800	3.372
1119	combined_composition_shift	Calibrated stack	15	0.000	0.088	0.467	0.933	3.617
	combined_composition_shift	Calibrated stack	15	0.000	0.106	0.467	0.867	2.893
1120	combined_composition_shift	Calibrated stack	15	0.133	0.085	0.267	0.800	4.394
1121	combined_composition_shift	Calibrated stack	15	0.133	0.073	0.333	0.800	3.717
	combined_composition_shift	Calibrated stack	15	0.067	0.092	0.667	0.800	2.792
1122	combined_composition_shift	Calibrated stack	15	0.067	0.069	0.267	0.933	3.909
1123	combined_composition_shift	Calibrated stack	15	0.133	0.059	0.267	0.867	3.564
	combined_composition_shift	Calibrated stack	15	0.067	0.070	0.200	0.867	3.742
1124	combined_composition_shift	Old EBM	15	0.000	0.089	0.400	0.933	3.905
1125	combined_composition_shift	Old EBM	15	0.000	0.109	0.400	0.867	3.340
	combined_composition_shift	Old EBM	15	0.067	0.088	0.333	0.800	4.580
1126	combined_composition_shift	Old EBM	15	0.000	0.094	0.533	0.867	3.179
1127	combined_composition_shift	Old EBM	15	0.067	0.117	0.600	0.867	3.022
	combined_composition_shift	Old EBM	15	0.067	0.072	0.333	0.867	3.713
1128	combined_composition_shift	Old EBM	15	0.133	0.059	0.267	0.867	4.489
1129	combined_composition_shift	Old EBM	15	0.067	0.079	0.400	0.933	3.078
	combined_composition_shift	EBM v5	15	0.000	0.093	0.533	0.933	3.230
1130	combined_composition_shift	EBM v5	15	0.000	0.087	0.400	1.000	3.384
1131	combined_composition_shift	EBM v5	15	0.067	0.088	0.267	0.867	4.237
	combined_composition_shift	EBM v5	15	0.133	0.080	0.267	0.800	3.836
1132	combined_composition_shift	EBM v5	15	0.067	0.102	0.667	0.800	2.885
1133	combined_composition_shift	EBM v5	15	0.067	0.089	0.467	0.867	3.448
	combined_composition_shift	EBM v5	15	0.133	0.069	0.467	0.867	3.383

	Split	Method	Episodes	Success	Slip	Drop	Collision	Energy
1134								
1135								
1136	combined_composition_shift	EBM v5	15	0.067	0.073	0.267	0.933	3.543
1137	combined_composition_shift	Oracle grid	15	0.133	0.045	0.000	0.800	5.614
1138	combined_composition_shift	Oracle grid	15	0.067	0.056	0.067	0.867	3.785
1139	combined_composition_shift	Oracle grid	15	0.333	0.039	0.000	0.667	4.812
1140	combined_composition_shift	Oracle grid	15	0.333	0.036	0.000	0.667	4.981
1141	combined_composition_shift	Oracle grid	15	0.133	0.078	0.133	0.733	3.191
1142	combined_composition_shift	Oracle grid	15	0.200	0.044	0.000	0.800	4.732
1143	combined_composition_shift	Oracle grid	15	0.400	0.035	0.067	0.533	5.101
1144	combined_composition_shift	Oracle grid	15	0.200	0.032	0.000	0.800	4.020
1145	long_bar_high_aspect	Random	15	0.200	0.070	0.733	0.000	1.151
1146	long_bar_high_aspect	Random	15	0.133	0.078	0.733	0.000	1.658
1147	long_bar_high_aspect	Random	15	0.133	0.126	0.733	0.000	1.105
1148	long_bar_high_aspect	Random	15	0.200	0.073	0.800	0.000	1.047
1149	long_bar_high_aspect	Random	15	0.067	0.121	0.933	0.000	0.839
1150	long_bar_high_aspect	Random	15	0.067	0.105	0.867	0.000	1.390
1151	long_bar_high_aspect	Random	15	0.200	0.091	0.800	0.000	0.966
1152	long_bar_high_aspect	Random	15	0.200	0.109	0.733	0.000	1.763
1153	long_bar_high_aspect	Antipodal	15	0.200	0.067	0.733	0.000	0.702
1154	long_bar_high_aspect	Antipodal	15	0.267	0.064	0.667	0.000	0.904
1155	long_bar_high_aspect	Antipodal	15	0.133	0.073	0.733	0.000	0.731
1156	long_bar_high_aspect	Antipodal	15	0.267	0.071	0.667	0.000	0.847
1157	long_bar_high_aspect	Antipodal	15	0.200	0.063	0.800	0.000	0.783
1158	long_bar_high_aspect	Antipodal	15	0.200	0.067	0.800	0.000	0.665
1159	long_bar_high_aspect	Antipodal	15	0.133	0.066	0.800	0.000	0.761
1160	long_bar_high_aspect	Antipodal	15	0.067	0.071	0.867	0.000	0.598
1161	long_bar_high_aspect	Force closure	15	0.267	0.061	0.667	0.000	0.813
1162	long_bar_high_aspect	Force closure	15	0.267	0.067	0.733	0.000	0.777
1163	long_bar_high_aspect	Force closure	15	0.267	0.064	0.667	0.000	0.814
1164	long_bar_high_aspect	Force closure	15	0.267	0.066	0.667	0.000	0.836
1165	long_bar_high_aspect	Force closure	15	0.200	0.069	0.800	0.000	0.807
1166	long_bar_high_aspect	Force closure	15	0.267	0.061	0.733	0.000	0.714
1167	long_bar_high_aspect	Force closure	15	0.400	0.052	0.600	0.000	0.944
1168	long_bar_high_aspect	Force closure	15	0.333	0.057	0.667	0.000	0.754
1169	long_bar_high_aspect	Collision-aware FC	15	0.067	0.063	0.933	0.000	0.669
1170	long_bar_high_aspect	Collision-aware FC	15	0.000	0.063	1.000	0.000	0.553
1171	long_bar_high_aspect	Collision-aware FC	15	0.067	0.064	0.933	0.000	0.647
1172	long_bar_high_aspect	Collision-aware FC	15	0.000	0.059	1.000	0.000	0.554
1173	long_bar_high_aspect	Collision-aware FC	15	0.133	0.069	0.867	0.000	0.713
1174	long_bar_high_aspect	Collision-aware FC	15	0.000	0.063	1.000	0.000	0.548
1175	long_bar_high_aspect	Collision-aware FC	15	0.133	0.061	0.867	0.000	0.741
1176	long_bar_high_aspect	Collision-aware FC	15	0.000	0.063	1.000	0.000	0.555
1177	long_bar_high_aspect	CEM analytic	15	0.400	0.060	0.533	0.000	0.774
1178	long_bar_high_aspect	CEM analytic	15	0.333	0.068	0.667	0.000	0.805
1179	long_bar_high_aspect	CEM analytic	15	0.333	0.067	0.600	0.000	0.806
1180	long_bar_high_aspect	CEM analytic	15	0.267	0.069	0.667	0.000	0.836
1181	long_bar_high_aspect	CEM analytic	15	0.267	0.064	0.733	0.000	0.865
1182	long_bar_high_aspect	CEM analytic	15	0.267	0.057	0.733	0.000	0.747
1183	long_bar_high_aspect	CEM analytic	15	0.400	0.058	0.533	0.000	0.962
1184	long_bar_high_aspect	CEM analytic	15	0.333	0.053	0.667	0.000	0.752
1185	long_bar_high_aspect	Robust perturb.	15	0.067	0.063	0.933	0.000	0.670
1186	long_bar_high_aspect	Robust perturb.	15	0.000	0.060	1.000	0.000	0.552
1187	long_bar_high_aspect	Robust perturb.	15	0.000	0.073	1.000	0.000	0.572
1188	long_bar_high_aspect	Robust perturb.	15	0.000	0.068	1.000	0.000	0.569
1189	long_bar_high_aspect	Robust perturb.	15	0.000	0.069	1.000	0.000	0.565
1190	long_bar_high_aspect	Robust perturb.	15	0.067	0.063	0.933	0.000	0.638
1191	long_bar_high_aspect	Robust perturb.	15	0.133	0.068	0.867	0.000	0.755
1192	long_bar_high_aspect	Robust perturb.	15	0.000	0.062	1.000	0.000	0.565
1193	long_bar_high_aspect	MLP energy	15	0.933	0.050	0.000	0.000	1.537
1194	long_bar_high_aspect	MLP energy	15	0.867	0.066	0.000	0.000	1.394
1195	long_bar_high_aspect	MLP energy	15	1.000	0.053	0.000	0.000	1.525
1196	long_bar_high_aspect	MLP energy	15	0.733	0.061	0.067	0.000	1.401
1197	long_bar_high_aspect	MLP energy	15	0.933	0.064	0.000	0.000	1.564
1198	long_bar_high_aspect	MLP energy	15	0.867	0.067	0.000	0.000	1.487
1199	long_bar_high_aspect	MLP energy	15	0.933	0.051	0.000	0.000	1.485
1200	long_bar_high_aspect	MLP energy	15	0.933	0.045	0.000	0.000	1.345
1201	long_bar_high_aspect	HGB energy	15	1.000	0.043	0.000	0.000	1.413
1202	long_bar_high_aspect	HGB energy	15	0.933	0.056	0.067	0.000	1.302
1203	long_bar_high_aspect	HGB energy	15	0.933	0.045	0.067	0.000	1.397
1204	long_bar_high_aspect	HGB energy	15	0.933	0.044	0.067	0.000	1.244
1205	long_bar_high_aspect	HGB energy	15	1.000	0.053	0.000	0.000	1.652
1206	long_bar_high_aspect	HGB energy	15	0.933	0.045	0.067	0.000	1.455
1207	long_bar_high_aspect	HGB energy	15	0.933	0.036	0.067	0.000	1.333
1208	long_bar_high_aspect	HGB energy	15	0.933	0.043	0.067	0.000	1.198
1209	long_bar_high_aspect	Transformer	15	1.000	0.044	0.000	0.000	1.540
1210	long_bar_high_aspect	Transformer	15	0.800	0.073	0.000	0.000	1.497
1211	long_bar_high_aspect	Transformer	15	0.933	0.052	0.000	0.000	1.701
1212	long_bar_high_aspect	Transformer	15	0.600	0.072	0.067	0.000	1.412
1213	long_bar_high_aspect	Transformer	15	0.933	0.054	0.000	0.000	1.689
1214	long_bar_high_aspect	Transformer	15	0.800	0.062	0.000	0.000	1.546

Split	Method	Episodes	Success	Slip	Drop	Collision	Energy
long_bar_high_aspect	Transformer	15	1.000	0.039	0.000	0.000	1.597
long_bar_high_aspect	Transformer	15	0.800	0.061	0.067	0.000	1.547
long_bar_high_aspect	Ensemble	15	1.000	0.041	0.000	0.000	1.381
long_bar_high_aspect	Ensemble	15	0.933	0.052	0.067	0.000	1.293
long_bar_high_aspect	Ensemble	15	1.000	0.044	0.000	0.000	1.212
long_bar_high_aspect	Ensemble	15	0.933	0.043	0.067	0.000	1.182
long_bar_high_aspect	Ensemble	15	1.000	0.060	0.000	0.000	1.575
long_bar_high_aspect	Ensemble	15	0.867	0.064	0.133	0.000	1.395
long_bar_high_aspect	Ensemble	15	0.933	0.056	0.067	0.000	1.562
long_bar_high_aspect	Ensemble	15	1.000	0.039	0.000	0.000	1.425
long_bar_high_aspect	Calibrated stack	15	1.000	0.049	0.000	0.000	1.618
long_bar_high_aspect	Calibrated stack	15	0.933	0.058	0.067	0.000	1.693
long_bar_high_aspect	Calibrated stack	15	1.000	0.051	0.000	0.000	1.764
long_bar_high_aspect	Calibrated stack	15	0.933	0.054	0.067	0.000	1.467
long_bar_high_aspect	Calibrated stack	15	1.000	0.056	0.000	0.000	1.779
long_bar_high_aspect	Calibrated stack	15	0.933	0.050	0.000	0.000	1.837
long_bar_high_aspect	Calibrated stack	15	1.000	0.043	0.000	0.000	1.807
long_bar_high_aspect	Calibrated stack	15	1.000	0.049	0.000	0.000	1.609
long_bar_high_aspect	Old EBM	15	0.933	0.038	0.000	0.000	1.556
long_bar_high_aspect	Old EBM	15	0.667	0.080	0.067	0.000	1.544
long_bar_high_aspect	Old EBM	15	0.867	0.049	0.067	0.000	1.643
long_bar_high_aspect	Old EBM	15	0.667	0.073	0.067	0.000	1.583
long_bar_high_aspect	Old EBM	15	0.867	0.064	0.000	0.000	1.771
long_bar_high_aspect	Old EBM	15	0.867	0.048	0.000	0.000	1.630
long_bar_high_aspect	Old EBM	15	1.000	0.041	0.000	0.000	1.716
long_bar_high_aspect	Old EBM	15	0.867	0.054	0.000	0.000	1.618
long_bar_high_aspect	EBM v5	15	0.933	0.050	0.000	0.000	1.689
long_bar_high_aspect	EBM v5	15	0.800	0.071	0.067	0.000	1.731
long_bar_high_aspect	EBM v5	15	0.800	0.058	0.133	0.000	1.606
long_bar_high_aspect	EBM v5	15	0.733	0.062	0.133	0.000	1.549
long_bar_high_aspect	EBM v5	15	0.933	0.055	0.000	0.000	1.794
long_bar_high_aspect	EBM v5	15	0.867	0.053	0.067	0.000	1.733
long_bar_high_aspect	EBM v5	15	1.000	0.046	0.000	0.000	1.819
long_bar_high_aspect	EBM v5	15	1.000	0.044	0.000	0.000	1.616
long_bar_high_aspect	Oracle grid	15	0.933	0.035	0.067	0.000	1.155
long_bar_high_aspect	Oracle grid	15	0.800	0.051	0.067	0.000	1.280
long_bar_high_aspect	Oracle grid	15	0.867	0.033	0.133	0.000	1.307
long_bar_high_aspect	Oracle grid	15	0.867	0.041	0.067	0.000	1.791
long_bar_high_aspect	Oracle grid	15	0.800	0.056	0.067	0.000	1.638
long_bar_high_aspect	Oracle grid	15	0.933	0.044	0.067	0.000	1.323
long_bar_high_aspect	Oracle grid	15	1.000	0.033	0.000	0.000	1.634
long_bar_high_aspect	Oracle grid	15	1.000	0.037	0.000	0.000	1.400
material_shift_proxy	Random	15	0.067	0.087	0.733	0.000	1.696
material_shift_proxy	Random	15	0.200	0.094	0.800	0.000	1.516
material_shift_proxy	Random	15	0.133	0.095	0.867	0.000	1.431
material_shift_proxy	Random	15	0.333	0.078	0.667	0.000	1.754
material_shift_proxy	Random	15	0.200	0.085	0.800	0.000	1.499
material_shift_proxy	Random	15	0.133	0.094	0.867	0.000	1.484
material_shift_proxy	Random	15	0.200	0.089	0.733	0.000	1.610
material_shift_proxy	Random	15	0.067	0.105	0.933	0.000	1.379
material_shift_proxy	Antipodal	15	0.467	0.090	0.533	0.000	2.053
material_shift_proxy	Antipodal	15	0.267	0.092	0.600	0.000	2.095
material_shift_proxy	Antipodal	15	0.200	0.089	0.733	0.000	1.898
material_shift_proxy	Antipodal	15	0.467	0.074	0.333	0.000	2.252
material_shift_proxy	Antipodal	15	0.133	0.092	0.667	0.000	2.262
material_shift_proxy	Antipodal	15	0.333	0.079	0.667	0.000	2.079
material_shift_proxy	Antipodal	15	0.533	0.084	0.333	0.000	2.350
material_shift_proxy	Antipodal	15	0.200	0.098	0.800	0.000	1.967
material_shift_proxy	Force closure	15	0.267	0.092	0.733	0.000	1.863
material_shift_proxy	Force closure	15	0.267	0.072	0.600	0.000	1.964
material_shift_proxy	Force closure	15	0.267	0.079	0.667	0.000	2.205
material_shift_proxy	Force closure	15	0.400	0.071	0.400	0.000	2.204
material_shift_proxy	Force closure	15	0.133	0.087	0.733	0.000	2.062
material_shift_proxy	Force closure	15	0.133	0.075	0.800	0.000	1.953
material_shift_proxy	Force closure	15	0.333	0.077	0.667	0.000	1.984
material_shift_proxy	Force closure	15	0.200	0.087	0.800	0.000	1.925
material_shift_proxy	Collision-aware FC	15	0.200	0.082	0.733	0.000	1.966
material_shift_proxy	Collision-aware FC	15	0.267	0.094	0.467	0.000	2.237
material_shift_proxy	Collision-aware FC	15	0.400	0.076	0.533	0.000	2.293
material_shift_proxy	Collision-aware FC	15	0.400	0.072	0.533	0.000	2.068
material_shift_proxy	Collision-aware FC	15	0.267	0.070	0.733	0.000	2.076
material_shift_proxy	Collision-aware FC	15	0.133	0.077	0.867	0.000	1.855
material_shift_proxy	Collision-aware FC	15	0.533	0.064	0.467	0.000	2.104
material_shift_proxy	Collision-aware FC	15	0.267	0.067	0.733	0.000	2.103
material_shift_proxy	CEM analytic	15	0.200	0.093	0.800	0.000	1.824
material_shift_proxy	CEM analytic	15	0.133	0.088	0.600	0.000	2.009
material_shift_proxy	CEM analytic	15	0.333	0.073	0.600	0.000	2.273
material_shift_proxy	CEM analytic	15	0.467	0.073	0.400	0.000	2.172
material_shift_proxy	CEM analytic	15	0.133	0.081	0.800	0.000	1.902

Split	Method	Episodes	Success	Slip	Drop	Collision	Energy
material_shift_proxy	CEM analytic	15	0.200	0.071	0.800	0.000	1.942
material_shift_proxy	CEM analytic	15	0.400	0.075	0.600	0.000	2.065
material_shift_proxy	CEM analytic	15	0.133	0.092	0.867	0.000	1.901
material_shift_proxy	Robust perturb.	15	0.133	0.081	0.667	0.000	2.134
material_shift_proxy	Robust perturb.	15	0.200	0.087	0.733	0.000	1.962
material_shift_proxy	Robust perturb.	15	0.267	0.085	0.733	0.000	2.110
material_shift_proxy	Robust perturb.	15	0.267	0.079	0.733	0.000	2.008
material_shift_proxy	Robust perturb.	15	0.400	0.056	0.600	0.000	2.252
material_shift_proxy	Robust perturb.	15	0.133	0.080	0.867	0.000	1.845
material_shift_proxy	Robust perturb.	15	0.267	0.084	0.733	0.000	1.938
material_shift_proxy	Robust perturb.	15	0.267	0.071	0.733	0.000	1.982
material_shift_proxy	MLP energy	15	0.267	0.082	0.733	0.000	1.748
material_shift_proxy	MLP energy	15	0.467	0.079	0.467	0.000	1.948
material_shift_proxy	MLP energy	15	0.533	0.076	0.333	0.000	2.011
material_shift_proxy	MLP energy	15	0.667	0.058	0.333	0.000	2.007
material_shift_proxy	MLP energy	15	0.400	0.076	0.600	0.000	1.721
material_shift_proxy	MLP energy	15	0.467	0.068	0.467	0.000	1.754
material_shift_proxy	MLP energy	15	0.333	0.064	0.600	0.000	1.883
material_shift_proxy	MLP energy	15	0.600	0.076	0.333	0.000	2.177
material_shift_proxy	HGB energy	15	0.733	0.066	0.133	0.000	2.096
material_shift_proxy	HGB energy	15	0.933	0.049	0.067	0.000	2.188
material_shift_proxy	HGB energy	15	0.667	0.070	0.200	0.000	1.912
material_shift_proxy	HGB energy	15	0.933	0.050	0.067	0.000	1.958
material_shift_proxy	HGB energy	15	0.867	0.064	0.133	0.000	2.042
material_shift_proxy	HGB energy	15	0.867	0.058	0.133	0.000	2.001
material_shift_proxy	HGB energy	15	0.800	0.062	0.133	0.000	2.005
material_shift_proxy	HGB energy	15	0.800	0.058	0.133	0.000	1.855
material_shift_proxy	Transformer	15	0.600	0.069	0.333	0.000	2.141
material_shift_proxy	Transformer	15	0.800	0.045	0.200	0.000	2.160
material_shift_proxy	Transformer	15	0.533	0.074	0.400	0.000	2.060
material_shift_proxy	Transformer	15	0.600	0.060	0.333	0.000	2.184
material_shift_proxy	Transformer	15	0.400	0.071	0.600	0.000	1.992
material_shift_proxy	Transformer	15	0.400	0.059	0.600	0.000	1.900
material_shift_proxy	Transformer	15	0.667	0.052	0.333	0.000	2.167
material_shift_proxy	Transformer	15	0.467	0.059	0.533	0.000	1.963
material_shift_proxy	Ensemble	15	0.933	0.047	0.067	0.000	1.954
material_shift_proxy	Ensemble	15	1.000	0.044	0.000	0.000	2.347
material_shift_proxy	Ensemble	15	0.733	0.063	0.067	0.000	2.005
material_shift_proxy	Ensemble	15	0.933	0.053	0.000	0.000	1.988
material_shift_proxy	Ensemble	15	0.733	0.066	0.200	0.000	1.937
material_shift_proxy	Ensemble	15	0.867	0.055	0.133	0.000	1.985
material_shift_proxy	Ensemble	15	0.800	0.064	0.000	0.000	2.107
material_shift_proxy	Ensemble	15	0.867	0.061	0.067	0.000	1.938
material_shift_proxy	Calibrated stack	15	0.533	0.071	0.467	0.000	2.058
material_shift_proxy	Calibrated stack	15	0.667	0.066	0.200	0.000	2.360
material_shift_proxy	Calibrated stack	15	0.733	0.068	0.200	0.000	2.220
material_shift_proxy	Calibrated stack	15	0.867	0.057	0.067	0.000	2.306
material_shift_proxy	Calibrated stack	15	0.600	0.059	0.400	0.000	1.991
material_shift_proxy	Calibrated stack	15	0.533	0.076	0.400	0.000	1.930
material_shift_proxy	Calibrated stack	15	0.733	0.059	0.133	0.000	2.291
material_shift_proxy	Calibrated stack	15	0.867	0.057	0.133	0.000	2.380
material_shift_proxy	Old EBM	15	0.333	0.075	0.600	0.000	2.120
material_shift_proxy	Old EBM	15	0.667	0.061	0.267	0.000	2.345
material_shift_proxy	Old EBM	15	0.400	0.064	0.600	0.000	2.080
material_shift_proxy	Old EBM	15	0.600	0.059	0.333	0.000	2.251
material_shift_proxy	Old EBM	15	0.333	0.069	0.667	0.000	2.107
material_shift_proxy	Old EBM	15	0.200	0.072	0.800	0.000	1.982
material_shift_proxy	Old EBM	15	0.267	0.055	0.733	0.000	1.986
material_shift_proxy	Old EBM	15	0.467	0.066	0.533	0.000	2.188
material_shift_proxy	EBM v5	15	0.400	0.071	0.600	0.000	2.334
material_shift_proxy	EBM v5	15	0.800	0.050	0.133	0.000	2.610
material_shift_proxy	EBM v5	15	0.600	0.066	0.400	0.000	2.324
material_shift_proxy	EBM v5	15	0.600	0.075	0.333	0.000	2.373
material_shift_proxy	EBM v5	15	0.267	0.064	0.733	0.000	2.137
material_shift_proxy	EBM v5	15	0.267	0.070	0.733	0.000	2.102
material_shift_proxy	EBM v5	15	0.400	0.070	0.600	0.000	2.192
material_shift_proxy	EBM v5	15	0.600	0.058	0.400	0.000	2.420
material_shift_proxy	Oracle grid	15	0.533	0.068	0.267	0.000	2.465
material_shift_proxy	Oracle grid	15	0.667	0.052	0.267	0.000	2.517
material_shift_proxy	Oracle grid	15	0.667	0.059	0.200	0.000	2.528
material_shift_proxy	Oracle grid	15	0.733	0.059	0.133	0.000	2.506
material_shift_proxy	Oracle grid	15	0.733	0.048	0.200	0.000	2.693
material_shift_proxy	Oracle grid	15	0.667	0.053	0.333	0.000	2.219
material_shift_proxy	Oracle grid	15	0.800	0.045	0.200	0.000	2.429
material_shift_proxy	Oracle grid	15	0.667	0.055	0.333	0.000	2.482
seen_simple	Random	15	0.933	0.054	0.000	0.000	2.276
seen_simple	Random	15	0.867	0.065	0.000	0.000	1.932
seen_simple	Random	15	0.933	0.064	0.000	0.000	1.918
seen_simple	Random	15	0.933	0.060	0.067	0.000	2.043

	Split	Method	Episodes	Success	Slip	Drop	Collision	Energy
1296	seen_simple	Random	15	0.867	0.063	0.000	0.000	1.742
1297	seen_simple	Random	15	0.800	0.056	0.067	0.000	2.053
1298	seen_simple	Random	15	1.000	0.067	0.000	0.000	1.998
1299	seen_simple	Random	15	0.600	0.090	0.000	0.000	2.280
1300	seen_simple	Antipodal	15	0.867	0.070	0.000	0.000	2.175
1301	seen_simple	Antipodal	15	0.800	0.062	0.000	0.000	2.462
1302	seen_simple	Antipodal	15	0.800	0.070	0.000	0.000	2.478
1303	seen_simple	Antipodal	15	0.733	0.079	0.000	0.000	2.478
1304	seen_simple	Antipodal	15	0.933	0.064	0.000	0.000	2.260
1305	seen_simple	Antipodal	15	0.800	0.062	0.000	0.000	2.470
1306	seen_simple	Antipodal	15	0.800	0.068	0.000	0.000	2.253
1307	seen_simple	Antipodal	15	0.733	0.072	0.000	0.000	2.382
1308	seen_simple	Force closure	15	0.933	0.062	0.000	0.000	2.414
1309	seen_simple	Force closure	15	0.867	0.054	0.000	0.000	2.442
1310	seen_simple	Force closure	15	0.933	0.051	0.000	0.000	2.544
1311	seen_simple	Force closure	15	0.933	0.052	0.000	0.000	2.482
1312	seen_simple	Force closure	15	0.933	0.063	0.000	0.000	2.289
1313	seen_simple	Force closure	15	0.867	0.046	0.000	0.000	2.381
1314	seen_simple	Force closure	15	0.867	0.054	0.000	0.000	2.314
1315	seen_simple	Force closure	15	0.867	0.055	0.000	0.000	2.340
1316	seen_simple	Collision-aware FC	15	1.000	0.049	0.000	0.000	2.933
1317	seen_simple	Collision-aware FC	15	0.867	0.066	0.000	0.000	2.726
1318	seen_simple	Collision-aware FC	15	0.867	0.051	0.000	0.000	2.700
1319	seen_simple	Collision-aware FC	15	0.867	0.062	0.000	0.000	2.638
1320	seen_simple	Collision-aware FC	15	1.000	0.066	0.000	0.000	2.643
1321	seen_simple	Collision-aware FC	15	0.867	0.051	0.000	0.000	2.562
1322	seen_simple	Collision-aware FC	15	0.933	0.047	0.000	0.000	2.516
1323	seen_simple	Collision-aware FC	15	1.000	0.040	0.000	0.000	2.650
1324	seen_simple	CEM analytic	15	0.933	0.061	0.000	0.000	2.572
1325	seen_simple	CEM analytic	15	0.867	0.056	0.000	0.000	2.711
1326	seen_simple	CEM analytic	15	0.867	0.061	0.000	0.000	2.737
1327	seen_simple	CEM analytic	15	0.800	0.061	0.000	0.000	2.693
1328	seen_simple	CEM analytic	15	0.933	0.062	0.000	0.000	2.422
1329	seen_simple	CEM analytic	15	0.867	0.051	0.000	0.000	2.614
1330	seen_simple	CEM analytic	15	0.800	0.049	0.000	0.000	2.322
1331	seen_simple	CEM analytic	15	0.933	0.045	0.000	0.000	2.532
1332	seen_simple	Robust perturb.	15	1.000	0.056	0.000	0.000	2.747
1333	seen_simple	Robust perturb.	15	0.867	0.068	0.000	0.000	2.730
1334	seen_simple	Robust perturb.	15	1.000	0.043	0.000	0.000	2.769
1335	seen_simple	Robust perturb.	15	0.800	0.066	0.000	0.000	3.046
1336	seen_simple	Robust perturb.	15	1.000	0.064	0.000	0.000	2.752
1337	seen_simple	Robust perturb.	15	0.933	0.051	0.000	0.000	2.673
1338	seen_simple	Robust perturb.	15	0.933	0.047	0.000	0.000	2.859
1339	seen_simple	Robust perturb.	15	0.933	0.050	0.000	0.000	2.590
1340	seen_simple	MLP energy	15	1.000	0.028	0.000	0.000	2.183
1341	seen_simple	MLP energy	15	1.000	0.038	0.000	0.000	2.067
1342	seen_simple	MLP energy	15	1.000	0.022	0.000	0.000	2.100
1343	seen_simple	MLP energy	15	1.000	0.037	0.000	0.000	2.070
1344	seen_simple	MLP energy	15	1.000	0.030	0.000	0.000	2.103
1345	seen_simple	MLP energy	15	1.000	0.029	0.000	0.000	2.327
1346	seen_simple	MLP energy	15	1.000	0.024	0.000	0.000	2.193
1347	seen_simple	MLP energy	15	1.000	0.027	0.000	0.000	2.099
1348	seen_simple	HGB energy	15	1.000	0.033	0.000	0.000	2.700
1349	seen_simple	HGB energy	15	1.000	0.039	0.000	0.000	2.221
1350	seen_simple	HGB energy	15	1.000	0.024	0.000	0.000	2.514
1351	seen_simple	HGB energy	15	1.000	0.035	0.000	0.000	2.442
1352	seen_simple	HGB energy	15	1.000	0.033	0.000	0.000	2.476
1353	seen_simple	HGB energy	15	1.000	0.035	0.000	0.000	2.637
1354	seen_simple	HGB energy	15	1.000	0.030	0.000	0.000	2.516
1355	seen_simple	HGB energy	15	1.000	0.043	0.000	0.000	2.339
1356	seen_simple	Transformer	15	1.000	0.028	0.000	0.000	2.022
1357	seen_simple	Transformer	15	1.000	0.022	0.000	0.000	2.101
1358	seen_simple	Transformer	15	1.000	0.042	0.000	0.000	2.248
1359	seen_simple	Transformer	15	1.000	0.029	0.000	0.000	1.949
1360	seen_simple	Transformer	15	1.000	0.032	0.000	0.000	2.085
1361	seen_simple	Transformer	15	0.933	0.042	0.000	0.000	2.071
1362	seen_simple	Transformer	15	0.933	0.039	0.000	0.000	2.282
1363	seen_simple	Transformer	15	1.000	0.046	0.000	0.000	2.141
1364	seen_simple	Ensemble	15	1.000	0.038	0.000	0.000	2.529
1365	seen_simple	Ensemble	15	1.000	0.044	0.000	0.000	2.461
1366	seen_simple	Ensemble	15	1.000	0.036	0.000	0.000	2.554
1367	seen_simple	Ensemble	15	1.000	0.038	0.000	0.000	2.423
1368	seen_simple	Ensemble	15	1.000	0.042	0.000	0.000	2.514
1369	seen_simple	Ensemble	15	1.000	0.035	0.000	0.000	2.674
1370	seen_simple	Ensemble	15	1.000	0.046	0.000	0.000	2.596
1371	seen_simple	Ensemble	15	1.000	0.048	0.000	0.000	2.481
1372	seen_simple	Calibrated stack	15	1.000	0.037	0.000	0.000	2.790
1373	seen_simple	Calibrated stack	15	1.000	0.032	0.000	0.000	2.548
1374	seen_simple	Calibrated stack	15	1.000	0.031	0.000	0.000	2.889

	Split	Method	Episodes	Success	Slip	Drop	Collision	Energy
1350								
1351	seen_simple	Calibrated stack	15	1.000	0.034	0.000	0.000	2.672
1352	seen_simple	Calibrated stack	15	1.000	0.031	0.000	0.000	2.808
1353	seen_simple	Calibrated stack	15	1.000	0.040	0.000	0.000	2.676
1354	seen_simple	Calibrated stack	15	1.000	0.038	0.000	0.000	2.936
	seen_simple	Calibrated stack	15	1.000	0.038	0.000	0.000	2.748
1355	seen_simple	Old EBM	15	1.000	0.043	0.000	0.000	2.606
	seen_simple	Old EBM	15	1.000	0.053	0.000	0.000	2.682
1356	seen_simple	Old EBM	15	1.000	0.045	0.000	0.000	2.791
	seen_simple	Old EBM	15	1.000	0.056	0.000	0.000	2.729
1357	seen_simple	Old EBM	15	0.933	0.049	0.000	0.000	2.213
1358	seen_simple	Old EBM	15	0.933	0.046	0.000	0.000	2.620
1359	seen_simple	Old EBM	15	1.000	0.042	0.000	0.000	2.404
	seen_simple	Old EBM	15	1.000	0.037	0.000	0.000	2.692
	seen_simple	EBM v5	15	1.000	0.039	0.000	0.000	2.844
1361	seen_simple	EBM v5	15	1.000	0.043	0.000	0.000	2.692
	seen_simple	EBM v5	15	1.000	0.036	0.000	0.000	2.569
1362	seen_simple	EBM v5	15	1.000	0.037	0.000	0.000	2.790
1363	seen_simple	EBM v5	15	1.000	0.044	0.000	0.000	2.800
	seen_simple	EBM v5	15	1.000	0.046	0.000	0.000	2.816
1364	seen_simple	EBM v5	15	1.000	0.042	0.000	0.000	2.720
1365	seen_simple	EBM v5	15	1.000	0.036	0.000	0.000	2.794
	seen_simple	Oracle grid	15	1.000	0.012	0.000	0.000	2.589
1366	seen_simple	Oracle grid	15	1.000	0.017	0.000	0.000	2.516
1367	seen_simple	Oracle grid	15	1.000	0.018	0.000	0.000	2.832
	seen_simple	Oracle grid	15	1.000	0.014	0.000	0.000	2.588
1368	seen_simple	Oracle grid	15	1.000	0.017	0.000	0.000	2.369
1369	seen_simple	Oracle grid	15	1.000	0.012	0.000	0.000	2.605
	seen_simple	Oracle grid	15	1.000	0.015	0.000	0.000	2.443
1370	seen_simple	Oracle grid	15	1.000	0.018	0.000	0.000	2.579
1371	slippery_contact	Random	15	0.267	0.104	0.667	0.000	1.555
1372	slippery_contact	Random	15	0.400	0.082	0.467	0.000	1.728
	slippery_contact	Random	15	0.400	0.086	0.600	0.000	1.704
1373	slippery_contact	Random	15	0.400	0.091	0.533	0.000	1.812
	slippery_contact	Random	15	0.333	0.102	0.533	0.000	1.767
1374	slippery_contact	Random	15	0.467	0.068	0.467	0.000	1.705
1375	slippery_contact	Random	15	0.600	0.079	0.400	0.000	1.905
	slippery_contact	Random	15	0.467	0.082	0.533	0.000	1.793
1376	slippery_contact	Antipodal	15	0.533	0.088	0.400	0.000	2.217
1377	slippery_contact	Antipodal	15	0.533	0.094	0.333	0.000	2.276
	slippery_contact	Antipodal	15	0.267	0.097	0.733	0.000	2.006
1378	slippery_contact	Antipodal	15	0.600	0.078	0.333	0.000	2.214
1379	slippery_contact	Antipodal	15	0.133	0.100	0.533	0.000	2.063
	slippery_contact	Antipodal	15	0.267	0.090	0.600	0.000	1.937
1380	slippery_contact	Antipodal	15	0.267	0.097	0.600	0.000	1.953
1381	slippery_contact	Antipodal	15	0.400	0.094	0.533	0.000	1.993
	slippery_contact	Force closure	15	0.467	0.083	0.467	0.000	2.207
1382	slippery_contact	Force closure	15	0.400	0.071	0.600	0.000	1.896
1383	slippery_contact	Force closure	15	0.133	0.091	0.867	0.000	1.749
	slippery_contact	Force closure	15	0.467	0.089	0.467	0.000	2.093
1384	slippery_contact	Force closure	15	0.200	0.077	0.733	0.000	1.935
1385	slippery_contact	Force closure	15	0.267	0.080	0.667	0.000	1.957
	slippery_contact	Force closure	15	0.333	0.076	0.600	0.000	2.066
1386	slippery_contact	Force closure	15	0.333	0.085	0.600	0.000	1.980
1387	slippery_contact	Collision-aware FC	15	0.467	0.075	0.533	0.000	2.216
	slippery_contact	Collision-aware FC	15	0.333	0.082	0.600	0.000	1.969
1388	slippery_contact	Collision-aware FC	15	0.200	0.099	0.800	0.000	1.919
1389	slippery_contact	Collision-aware FC	15	0.333	0.080	0.667	0.000	1.905
	slippery_contact	Collision-aware FC	15	0.333	0.085	0.600	0.000	2.153
1390	slippery_contact	Collision-aware FC	15	0.267	0.075	0.667	0.000	2.058
1391	slippery_contact	Collision-aware FC	15	0.267	0.066	0.733	0.000	2.007
	slippery_contact	Collision-aware FC	15	0.400	0.075	0.533	0.000	2.149
1392	slippery_contact	CEM analytic	15	0.333	0.082	0.600	0.000	2.129
1393	slippery_contact	CEM analytic	15	0.400	0.070	0.600	0.000	2.005
	slippery_contact	CEM analytic	15	0.133	0.096	0.867	0.000	1.828
1394	slippery_contact	CEM analytic	15	0.467	0.084	0.533	0.000	1.999
1395	slippery_contact	CEM analytic	15	0.067	0.098	0.800	0.000	1.943
	slippery_contact	CEM analytic	15	0.400	0.071	0.467	0.000	2.187
1396	slippery_contact	CEM analytic	15	0.333	0.074	0.667	0.000	1.989
1397	slippery_contact	CEM analytic	15	0.333	0.082	0.533	0.000	2.138
	slippery_contact	Robust perturb.	15	0.400	0.090	0.600	0.000	2.142
1398	slippery_contact	Robust perturb.	15	0.267	0.084	0.600	0.000	2.107
1399	slippery_contact	Robust perturb.	15	0.133	0.095	0.867	0.000	2.009
	slippery_contact	Robust perturb.	15	0.200	0.075	0.800	0.000	1.766
1400	slippery_contact	Robust perturb.	15	0.333	0.088	0.667	0.000	2.106
1401	slippery_contact	Robust perturb.	15	0.400	0.075	0.533	0.000	2.079
	slippery_contact	Robust perturb.	15	0.267	0.088	0.733	0.000	2.016
1402	slippery_contact	Robust perturb.	15	0.467	0.069	0.467	0.000	2.238
1403	slippery_contact	MLP energy	15	0.800	0.052	0.133	0.000	1.923
	slippery_contact	MLP energy	15	0.867	0.054	0.133	0.000	1.946

	Split	Method	Episodes	Success	Slip	Drop	Collision	Energy
1404								
1405								
1406	slippery_contact	MLP energy	15	0.800	0.056	0.200	0.000	1.890
1407	slippery_contact	MLP energy	15	0.933	0.053	0.067	0.000	2.013
1408	slippery_contact	MLP energy	15	0.867	0.053	0.133	0.000	2.153
1409	slippery_contact	MLP energy	15	1.000	0.054	0.000	0.000	2.021
1410	slippery_contact	MLP energy	15	0.733	0.052	0.200	0.000	1.829
1411	slippery_contact	MLP energy	15	0.867	0.055	0.133	0.000	1.847
1412	slippery_contact	HGB energy	15	1.000	0.048	0.000	0.000	2.180
1413	slippery_contact	HGB energy	15	1.000	0.055	0.000	0.000	2.020
1414	slippery_contact	HGB energy	15	0.933	0.055	0.067	0.000	2.097
1415	slippery_contact	HGB energy	15	1.000	0.053	0.000	0.000	1.905
1416	slippery_contact	HGB energy	15	0.933	0.047	0.067	0.000	2.209
1417	slippery_contact	HGB energy	15	0.867	0.063	0.067	0.000	2.140
1418	slippery_contact	HGB energy	15	0.933	0.058	0.000	0.000	2.037
1419	slippery_contact	HGB energy	15	0.933	0.052	0.067	0.000	2.040
1420	slippery_contact	Transformer	15	0.933	0.052	0.067	0.000	1.839
1421	slippery_contact	Transformer	15	0.867	0.057	0.067	0.000	1.804
1422	slippery_contact	Transformer	15	0.933	0.050	0.067	0.000	2.053
1423	slippery_contact	Transformer	15	0.867	0.043	0.133	0.000	1.961
1424	slippery_contact	Transformer	15	0.933	0.053	0.067	0.000	2.089
1425	slippery_contact	Transformer	15	0.867	0.047	0.133	0.000	1.860
1426	slippery_contact	Transformer	15	0.733	0.064	0.200	0.000	1.933
1427	slippery_contact	Transformer	15	0.933	0.050	0.067	0.000	1.891
1428	slippery_contact	Ensemble	15	0.933	0.051	0.000	0.000	1.980
1429	slippery_contact	Ensemble	15	0.933	0.067	0.000	0.000	1.919
1430	slippery_contact	Ensemble	15	0.933	0.063	0.000	0.000	2.011
1431	slippery_contact	Ensemble	15	1.000	0.050	0.000	0.000	1.958
1432	slippery_contact	Ensemble	15	0.933	0.062	0.067	0.000	2.162
1433	slippery_contact	Ensemble	15	0.933	0.057	0.000	0.000	2.078
1434	slippery_contact	Ensemble	15	0.867	0.064	0.067	0.000	1.890
1435	slippery_contact	Ensemble	15	0.933	0.055	0.067	0.000	1.909
1436	slippery_contact	Calibrated stack	15	1.000	0.043	0.000	0.000	2.104
1437	slippery_contact	Calibrated stack	15	0.933	0.061	0.067	0.000	2.013
1438	slippery_contact	Calibrated stack	15	1.000	0.051	0.000	0.000	2.167
1439	slippery_contact	Calibrated stack	15	1.000	0.047	0.000	0.000	2.059
1440	slippery_contact	Calibrated stack	15	0.933	0.045	0.067	0.000	2.301
1441	slippery_contact	Calibrated stack	15	0.867	0.054	0.133	0.000	2.014
1442	slippery_contact	Calibrated stack	15	0.867	0.057	0.133	0.000	2.022
1443	slippery_contact	Calibrated stack	15	0.933	0.055	0.067	0.000	2.094
1444	slippery_contact	Old EBM	15	0.867	0.058	0.133	0.000	2.046
1445	slippery_contact	Old EBM	15	0.667	0.059	0.333	0.000	2.129
1446	slippery_contact	Old EBM	15	0.667	0.058	0.333	0.000	2.064
1447	slippery_contact	Old EBM	15	0.867	0.044	0.133	0.000	2.141
1448	slippery_contact	Old EBM	15	0.733	0.062	0.133	0.000	2.174
1449	slippery_contact	Old EBM	15	0.533	0.062	0.400	0.000	1.944
1450	slippery_contact	Old EBM	15	0.467	0.072	0.400	0.000	2.142
1451	slippery_contact	Old EBM	15	0.667	0.058	0.333	0.000	2.178
1452	slippery_contact	EBM v5	15	0.667	0.060	0.333	0.000	2.277
1453	slippery_contact	EBM v5	15	0.733	0.068	0.267	0.000	2.288
1454	slippery_contact	EBM v5	15	0.533	0.059	0.467	0.000	2.021
1455	slippery_contact	EBM v5	15	0.533	0.067	0.467	0.000	2.194
1456	slippery_contact	EBM v5	15	0.667	0.067	0.267	0.000	2.313
1457	slippery_contact	EBM v5	15	0.400	0.082	0.467	0.000	2.045
	slippery_contact	EBM v5	15	0.533	0.069	0.467	0.000	2.059
	slippery_contact	EBM v5	15	0.600	0.055	0.400	0.000	2.156
	slippery_contact	Oracle grid	15	0.733	0.059	0.200	0.000	2.507
	slippery_contact	Oracle grid	15	0.933	0.040	0.067	0.000	2.553
	slippery_contact	Oracle grid	15	0.733	0.061	0.200	0.000	2.478
	slippery_contact	Oracle grid	15	0.867	0.053	0.133	0.000	2.317
	slippery_contact	Oracle grid	15	0.867	0.056	0.133	0.000	2.437
	slippery_contact	Oracle grid	15	0.933	0.046	0.067	0.000	2.647
	slippery_contact	Oracle grid	15	0.800	0.066	0.067	0.000	2.558
	slippery_contact	Oracle grid	15	0.933	0.045	0.067	0.000	2.567
	task_constraint_shift	Random	15	0.267	0.065	0.667	0.000	0.790
	task_constraint_shift	Random	15	0.267	0.064	0.467	0.000	2.182
	task_constraint_shift	Random	15	0.067	0.073	0.667	0.000	1.958
	task_constraint_shift	Random	15	0.200	0.080	0.667	0.000	2.242
	task_constraint_shift	Random	15	0.200	0.072	0.733	0.000	0.859
	task_constraint_shift	Random	15	0.200	0.060	0.667	0.000	1.730
	task_constraint_shift	Random	15	0.200	0.088	0.600	0.000	2.162
	task_constraint_shift	Random	15	0.067	0.083	0.733	0.000	1.270
	task_constraint_shift	Antipodal	15	0.067	0.075	0.667	0.000	1.781
	task_constraint_shift	Antipodal	15	0.067	0.058	0.533	0.000	2.609
	task_constraint_shift	Antipodal	15	0.067	0.067	0.600	0.000	2.518
	task_constraint_shift	Antipodal	15	0.000	0.064	0.867	0.000	1.298
	task_constraint_shift	Antipodal	15	0.067	0.061	0.867	0.000	0.975
	task_constraint_shift	Antipodal	15	0.133	0.054	0.533	0.000	2.541
	task_constraint_shift	Antipodal	15	0.000	0.068	0.667	0.000	2.443
	task_constraint_shift	Antipodal	15	0.133	0.061	0.667	0.000	1.912
	task_constraint_shift	Force closure	15	0.200	0.069	0.800	0.000	0.729

	Split	Method	Episodes	Success	Slip	Drop	Collision	Energy
1458								
1459	task_constraint_shift	Force closure	15	0.200	0.070	0.733	0.000	0.826
1460	task_constraint_shift	Force closure	15	0.200	0.058	0.800	0.000	0.758
1461	task_constraint_shift	Force closure	15	0.267	0.063	0.733	0.000	0.706
1462	task_constraint_shift	Force closure	15	0.133	0.060	0.867	0.000	0.638
1463	task_constraint_shift	Force closure	15	0.400	0.060	0.600	0.000	0.966
1464	task_constraint_shift	Force closure	15	0.333	0.066	0.533	0.000	1.260
1465	task_constraint_shift	Force closure	15	0.267	0.061	0.733	0.000	0.814
1466	task_constraint_shift	Collision-aware FC	15	0.133	0.068	0.867	0.000	0.703
1467	task_constraint_shift	Collision-aware FC	15	0.200	0.067	0.800	0.000	0.877
1468	task_constraint_shift	Collision-aware FC	15	0.000	0.071	1.000	0.000	0.570
1469	task_constraint_shift	Collision-aware FC	15	0.200	0.061	0.800	0.000	0.757
1470	task_constraint_shift	Collision-aware FC	15	0.067	0.068	0.933	0.000	0.665
1471	task_constraint_shift	Collision-aware FC	15	0.200	0.058	0.800	0.000	0.792
1472	task_constraint_shift	Collision-aware FC	15	0.267	0.068	0.733	0.000	1.233
1473	task_constraint_shift	Collision-aware FC	15	0.067	0.062	0.933	0.000	0.599
1474	task_constraint_shift	CEM analytic	15	0.200	0.069	0.800	0.000	0.729
1475	task_constraint_shift	CEM analytic	15	0.333	0.065	0.600	0.000	1.017
1476	task_constraint_shift	CEM analytic	15	0.333	0.059	0.667	0.000	0.843
1477	task_constraint_shift	CEM analytic	15	0.267	0.060	0.667	0.000	0.912
1478	task_constraint_shift	CEM analytic	15	0.200	0.061	0.800	0.000	0.767
1479	task_constraint_shift	CEM analytic	15	0.333	0.062	0.667	0.000	0.881
1480	task_constraint_shift	CEM analytic	15	0.400	0.068	0.467	0.000	1.352
1481	task_constraint_shift	CEM analytic	15	0.400	0.055	0.600	0.000	0.877
1482	task_constraint_shift	Robust perturb.	15	0.067	0.069	0.933	0.000	0.703
1483	task_constraint_shift	Robust perturb.	15	0.200	0.071	0.800	0.000	0.882
1484	task_constraint_shift	Robust perturb.	15	0.000	0.069	1.000	0.000	0.570
1485	task_constraint_shift	Robust perturb.	15	0.000	0.065	1.000	0.000	0.567
1486	task_constraint_shift	Robust perturb.	15	0.067	0.075	0.933	0.000	0.586
1487	task_constraint_shift	Robust perturb.	15	0.200	0.059	0.800	0.000	0.792
1488	task_constraint_shift	Robust perturb.	15	0.267	0.072	0.733	0.000	1.232
1489	task_constraint_shift	Robust perturb.	15	0.067	0.066	0.933	0.000	0.596
1490	task_constraint_shift	MLP energy	15	1.000	0.045	0.000	0.000	1.638
1491	task_constraint_shift	MLP energy	15	1.000	0.041	0.000	0.000	1.632
1492	task_constraint_shift	MLP energy	15	0.933	0.047	0.000	0.000	1.518
1493	task_constraint_shift	MLP energy	15	1.000	0.035	0.000	0.000	1.570
1494	task_constraint_shift	MLP energy	15	1.000	0.036	0.000	0.000	1.501
1495	task_constraint_shift	MLP energy	15	1.000	0.035	0.000	0.000	1.772
1496	task_constraint_shift	MLP energy	15	1.000	0.032	0.000	0.000	1.584
1497	task_constraint_shift	MLP energy	15	0.800	0.051	0.000	0.000	1.557
1498	task_constraint_shift	HGB energy	15	1.000	0.050	0.000	0.000	1.516
1499	task_constraint_shift	HGB energy	15	1.000	0.046	0.000	0.000	1.652
1500	task_constraint_shift	HGB energy	15	1.000	0.035	0.000	0.000	1.459
1501	task_constraint_shift	HGB energy	15	0.933	0.047	0.067	0.000	1.611
1502	task_constraint_shift	HGB energy	15	1.000	0.030	0.000	0.000	1.586
1503	task_constraint_shift	HGB energy	15	1.000	0.037	0.000	0.000	1.667
1504	task_constraint_shift	HGB energy	15	1.000	0.033	0.000	0.000	1.614
1505	task_constraint_shift	HGB energy	15	1.000	0.041	0.000	0.000	1.494
1506	task_constraint_shift	Transformer	15	1.000	0.043	0.000	0.000	1.772
1507	task_constraint_shift	Transformer	15	0.867	0.048	0.000	0.000	1.701
1508	task_constraint_shift	Transformer	15	0.867	0.054	0.000	0.000	1.524
1509	task_constraint_shift	Transformer	15	0.933	0.042	0.000	0.000	1.720
1510	task_constraint_shift	Transformer	15	1.000	0.040	0.000	0.000	1.761
1511	task_constraint_shift	Transformer	15	1.000	0.035	0.000	0.000	1.689
	task_constraint_shift	Transformer	15	0.933	0.044	0.000	0.000	1.809
	task_constraint_shift	Transformer	15	0.867	0.054	0.000	0.000	1.680
	task_constraint_shift	Ensemble	15	1.000	0.050	0.000	0.000	1.617
	task_constraint_shift	Ensemble	15	1.000	0.055	0.000	0.000	1.637
	task_constraint_shift	Ensemble	15	1.000	0.045	0.000	0.000	1.477
	task_constraint_shift	Ensemble	15	1.000	0.054	0.000	0.000	1.616
	task_constraint_shift	Ensemble	15	1.000	0.047	0.000	0.000	1.682
	task_constraint_shift	Ensemble	15	1.000	0.041	0.000	0.000	1.603
	task_constraint_shift	Ensemble	15	1.000	0.038	0.000	0.000	1.691
	task_constraint_shift	Ensemble	15	1.000	0.046	0.000	0.000	1.679
	task_constraint_shift	Calibrated stack	15	1.000	0.058	0.000	0.000	1.959
	task_constraint_shift	Calibrated stack	15	1.000	0.049	0.000	0.000	1.841
	task_constraint_shift	Calibrated stack	15	1.000	0.040	0.000	0.000	1.613
	task_constraint_shift	Calibrated stack	15	1.000	0.047	0.000	0.000	1.778
	task_constraint_shift	Calibrated stack	15	1.000	0.048	0.000	0.000	1.852
	task_constraint_shift	Calibrated stack	15	1.000	0.041	0.000	0.000	1.853
	task_constraint_shift	Calibrated stack	15	1.000	0.045	0.000	0.000	1.969
	task_constraint_shift	Calibrated stack	15	1.000	0.047	0.000	0.000	1.837
	task_constraint_shift	Old EBM	15	1.000	0.062	0.000	0.000	1.833
	task_constraint_shift	Old EBM	15	0.933	0.054	0.000	0.000	1.826
	task_constraint_shift	Old EBM	15	1.000	0.047	0.000	0.000	1.534
	task_constraint_shift	Old EBM	15	0.867	0.053	0.000	0.000	1.709
	task_constraint_shift	Old EBM	15	1.000	0.050	0.000	0.000	1.767
	task_constraint_shift	Old EBM	15	0.933	0.051	0.000	0.000	1.683
	task_constraint_shift	Old EBM	15	0.933	0.055	0.000	0.000	1.866
	task_constraint_shift	Old EBM	15	0.867	0.055	0.000	0.000	1.466

Split	Method	Episodes	Success	Slip	Drop	Collision	Energy
task_constraint_shift	EBM v5	15	1.000	0.060	0.000	0.000	1.822
task_constraint_shift	EBM v5	15	1.000	0.052	0.000	0.000	1.826
task_constraint_shift	EBM v5	15	1.000	0.045	0.000	0.000	1.527
task_constraint_shift	EBM v5	15	0.933	0.051	0.000	0.000	1.728
task_constraint_shift	EBM v5	15	1.000	0.048	0.000	0.000	1.788
task_constraint_shift	EBM v5	15	1.000	0.039	0.000	0.000	1.785
task_constraint_shift	EBM v5	15	1.000	0.054	0.000	0.000	1.962
task_constraint_shift	EBM v5	15	1.000	0.039	0.000	0.000	1.573
task_constraint_shift	Oracle grid	15	0.800	0.053	0.133	0.000	1.729
task_constraint_shift	Oracle grid	15	0.933	0.048	0.000	0.000	1.733
task_constraint_shift	Oracle grid	15	0.867	0.041	0.067	0.000	2.084
task_constraint_shift	Oracle grid	15	0.933	0.034	0.067	0.000	1.133
task_constraint_shift	Oracle grid	15	0.933	0.041	0.067	0.000	1.291
task_constraint_shift	Oracle grid	15	0.933	0.032	0.067	0.000	1.490
task_constraint_shift	Oracle grid	15	0.933	0.031	0.067	0.000	1.774
task_constraint_shift	Oracle grid	15	0.867	0.037	0.067	0.000	1.713
thin_handle_shift	Random	15	0.200	0.086	0.533	0.000	1.611
thin_handle_shift	Random	15	0.267	0.081	0.600	0.000	1.125
thin_handle_shift	Random	15	0.533	0.074	0.400	0.000	1.290
thin_handle_shift	Random	15	0.400	0.083	0.467	0.000	1.382
thin_handle_shift	Random	15	0.267	0.077	0.533	0.000	1.615
thin_handle_shift	Random	15	0.467	0.062	0.533	0.000	1.076
thin_handle_shift	Random	15	0.400	0.082	0.600	0.000	1.117
thin_handle_shift	Random	15	0.200	0.076	0.600	0.000	1.726
thin_handle_shift	Antipodal	15	0.000	0.079	0.267	0.000	3.361
thin_handle_shift	Antipodal	15	0.000	0.063	0.200	0.000	3.411
thin_handle_shift	Antipodal	15	0.000	0.077	0.533	0.000	2.526
thin_handle_shift	Antipodal	15	0.000	0.082	0.400	0.000	3.067
thin_handle_shift	Antipodal	15	0.000	0.084	0.400	0.000	2.945
thin_handle_shift	Antipodal	15	0.000	0.077	0.267	0.000	3.442
thin_handle_shift	Antipodal	15	0.000	0.054	0.133	0.000	4.171
thin_handle_shift	Antipodal	15	0.000	0.086	0.467	0.000	2.853
thin_handle_shift	Force closure	15	0.933	0.053	0.000	0.000	2.195
thin_handle_shift	Force closure	15	0.867	0.050	0.000	0.000	1.931
thin_handle_shift	Force closure	15	0.933	0.060	0.000	0.000	2.203
thin_handle_shift	Force closure	15	0.867	0.045	0.000	0.000	2.045
thin_handle_shift	Force closure	15	0.933	0.048	0.000	0.000	1.991
thin_handle_shift	Force closure	15	0.800	0.060	0.000	0.000	1.980
thin_handle_shift	Force closure	15	0.867	0.061	0.067	0.000	1.807
thin_handle_shift	Force closure	15	0.933	0.045	0.000	0.000	2.096
thin_handle_shift	Collision-aware FC	15	0.933	0.058	0.000	0.000	2.193
thin_handle_shift	Collision-aware FC	15	1.000	0.030	0.000	0.000	2.072
thin_handle_shift	Collision-aware FC	15	0.933	0.065	0.000	0.000	2.241
thin_handle_shift	Collision-aware FC	15	0.733	0.062	0.000	0.000	2.074
thin_handle_shift	Collision-aware FC	15	1.000	0.052	0.000	0.000	2.112
thin_handle_shift	Collision-aware FC	15	0.800	0.055	0.000	0.000	2.243
thin_handle_shift	Collision-aware FC	15	0.867	0.058	0.000	0.000	2.051
thin_handle_shift	Collision-aware FC	15	0.933	0.038	0.000	0.000	2.177
thin_handle_shift	CEM analytic	15	0.933	0.051	0.000	0.000	2.193
thin_handle_shift	CEM analytic	15	0.867	0.049	0.000	0.000	2.050
thin_handle_shift	CEM analytic	15	0.933	0.056	0.000	0.000	2.251
thin_handle_shift	CEM analytic	15	0.733	0.062	0.000	0.000	2.092
thin_handle_shift	CEM analytic	15	1.000	0.056	0.000	0.000	2.154
thin_handle_shift	CEM analytic	15	0.733	0.064	0.000	0.000	2.248
thin_handle_shift	CEM analytic	15	0.933	0.054	0.000	0.000	2.114
thin_handle_shift	CEM analytic	15	0.933	0.052	0.000	0.000	2.236
thin_handle_shift	Robust perturb.	15	0.867	0.065	0.000	0.000	2.182
thin_handle_shift	Robust perturb.	15	0.933	0.039	0.000	0.000	2.286
thin_handle_shift	Robust perturb.	15	0.933	0.062	0.000	0.000	2.244
thin_handle_shift	Robust perturb.	15	0.733	0.066	0.000	0.000	2.052
thin_handle_shift	Robust perturb.	15	0.867	0.066	0.000	0.000	2.121
thin_handle_shift	Robust perturb.	15	0.733	0.061	0.000	0.000	2.246
thin_handle_shift	Robust perturb.	15	0.933	0.057	0.000	0.000	2.206
thin_handle_shift	Robust perturb.	15	0.933	0.036	0.000	0.000	2.241
thin_handle_shift	MLP energy	15	1.000	0.042	0.000	0.000	1.886
thin_handle_shift	MLP energy	15	0.933	0.036	0.000	0.000	1.962
thin_handle_shift	MLP energy	15	1.000	0.046	0.000	0.000	1.820
thin_handle_shift	MLP energy	15	1.000	0.041	0.000	0.000	1.831
thin_handle_shift	MLP energy	15	1.000	0.049	0.000	0.000	1.935
thin_handle_shift	MLP energy	15	1.000	0.053	0.000	0.000	1.735
thin_handle_shift	MLP energy	15	0.933	0.036	0.000	0.000	1.723
thin_handle_shift	MLP energy	15	0.933	0.051	0.000	0.000	1.783
thin_handle_shift	HGB energy	15	1.000	0.036	0.000	0.000	2.086
thin_handle_shift	HGB energy	15	1.000	0.053	0.000	0.000	1.909
thin_handle_shift	HGB energy	15	1.000	0.047	0.000	0.000	2.070
thin_handle_shift	HGB energy	15	1.000	0.035	0.000	0.000	2.014
thin_handle_shift	HGB energy	15	1.000	0.049	0.000	0.000	1.980
thin_handle_shift	HGB energy	15	1.000	0.042	0.000	0.000	2.051
thin_handle_shift	HGB energy	15	1.000	0.053	0.000	0.000	2.079

Split	Method	Episodes	Success	Slip	Drop	Collision	Energy
thin_handle_shift	HGB energy	15	1.000	0.036	0.000	0.000	2.101
thin_handle_shift	Transformer	15	1.000	0.040	0.000	0.000	1.983
thin_handle_shift	Transformer	15	0.933	0.046	0.000	0.000	2.138
thin_handle_shift	Transformer	15	0.933	0.050	0.000	0.000	2.103
thin_handle_shift	Transformer	15	0.933	0.039	0.000	0.000	2.079
thin_handle_shift	Transformer	15	0.933	0.048	0.000	0.000	2.054
thin_handle_shift	Transformer	15	0.933	0.063	0.000	0.000	2.120
thin_handle_shift	Transformer	15	0.867	0.051	0.000	0.000	2.000
thin_handle_shift	Transformer	15	0.933	0.042	0.000	0.000	2.168
thin_handle_shift	Ensemble	15	1.000	0.050	0.000	0.000	2.269
thin_handle_shift	Ensemble	15	0.933	0.044	0.000	0.000	2.188
thin_handle_shift	Ensemble	15	1.000	0.051	0.000	0.000	2.257
thin_handle_shift	Ensemble	15	1.000	0.041	0.000	0.000	2.046
thin_handle_shift	Ensemble	15	1.000	0.045	0.000	0.000	2.042
thin_handle_shift	Ensemble	15	1.000	0.041	0.000	0.000	2.104
thin_handle_shift	Ensemble	15	1.000	0.047	0.000	0.000	2.146
thin_handle_shift	Ensemble	15	1.000	0.048	0.000	0.000	1.992
thin_handle_shift	Calibrated stack	15	1.000	0.047	0.000	0.000	2.221
thin_handle_shift	Calibrated stack	15	1.000	0.039	0.000	0.000	2.223
thin_handle_shift	Calibrated stack	15	1.000	0.057	0.000	0.000	2.215
thin_handle_shift	Calibrated stack	15	1.000	0.038	0.000	0.000	2.153
thin_handle_shift	Calibrated stack	15	1.000	0.051	0.000	0.000	2.145
thin_handle_shift	Calibrated stack	15	0.933	0.044	0.000	0.000	2.178
thin_handle_shift	Calibrated stack	15	1.000	0.042	0.000	0.000	2.186
thin_handle_shift	Calibrated stack	15	1.000	0.035	0.000	0.000	2.183
thin_handle_shift	Old EBM	15	0.933	0.051	0.000	0.000	2.195
thin_handle_shift	Old EBM	15	0.933	0.045	0.000	0.000	2.136
thin_handle_shift	Old EBM	15	0.933	0.051	0.000	0.000	2.229
thin_handle_shift	Old EBM	15	0.867	0.056	0.000	0.000	2.119
thin_handle_shift	Old EBM	15	1.000	0.056	0.000	0.000	2.143
thin_handle_shift	Old EBM	15	0.800	0.061	0.000	0.000	2.300
thin_handle_shift	Old EBM	15	0.933	0.057	0.000	0.000	2.229
thin_handle_shift	Old EBM	15	0.933	0.048	0.000	0.000	2.259
thin_handle_shift	EBM v5	15	1.000	0.054	0.000	0.000	2.234
thin_handle_shift	EBM v5	15	1.000	0.036	0.000	0.000	2.227
thin_handle_shift	EBM v5	15	1.000	0.056	0.000	0.000	2.222
thin_handle_shift	EBM v5	15	1.000	0.041	0.000	0.000	2.077
thin_handle_shift	EBM v5	15	1.000	0.055	0.000	0.000	2.161
thin_handle_shift	EBM v5	15	0.867	0.052	0.000	0.000	2.212
thin_handle_shift	EBM v5	15	1.000	0.049	0.000	0.000	2.260
thin_handle_shift	EBM v5	15	0.933	0.036	0.000	0.000	2.237
thin_handle_shift	Oracle grid	15	1.000	0.018	0.000	0.000	1.623
thin_handle_shift	Oracle grid	15	1.000	0.014	0.000	0.000	1.667
thin_handle_shift	Oracle grid	15	1.000	0.023	0.000	0.000	1.670
thin_handle_shift	Oracle grid	15	1.000	0.024	0.000	0.000	1.522
thin_handle_shift	Oracle grid	15	1.000	0.019	0.000	0.000	1.546
thin_handle_shift	Oracle grid	15	1.000	0.023	0.000	0.000	1.621
thin_handle_shift	Oracle grid	15	1.000	0.021	0.000	0.000	1.482
thin_handle_shift	Oracle grid	15	1.000	0.022	0.000	0.000	1.883
unseen_dimensions	Random	15	0.333	0.079	0.533	0.000	1.492
unseen_dimensions	Random	15	0.400	0.070	0.533	0.000	1.909
unseen_dimensions	Random	15	0.400	0.061	0.600	0.000	1.745
unseen_dimensions	Random	15	0.467	0.058	0.400	0.000	1.180
unseen_dimensions	Random	15	0.267	0.082	0.667	0.000	1.482
unseen_dimensions	Random	15	0.533	0.070	0.467	0.000	1.645
unseen_dimensions	Random	15	0.400	0.067	0.533	0.000	1.532
unseen_dimensions	Random	15	0.333	0.065	0.667	0.000	1.443
unseen_dimensions	Antipodal	15	0.800	0.061	0.133	0.000	3.886
unseen_dimensions	Antipodal	15	0.667	0.065	0.067	0.000	4.369
unseen_dimensions	Antipodal	15	0.800	0.064	0.067	0.000	4.323
unseen_dimensions	Antipodal	15	0.800	0.057	0.000	0.000	4.509
unseen_dimensions	Antipodal	15	0.800	0.062	0.067	0.000	3.968
unseen_dimensions	Antipodal	15	0.733	0.069	0.067	0.000	4.144
unseen_dimensions	Antipodal	15	0.667	0.064	0.133	0.000	4.099
unseen_dimensions	Antipodal	15	0.600	0.067	0.200	0.000	3.578
unseen_dimensions	Force closure	15	0.600	0.064	0.333	0.000	3.494
unseen_dimensions	Force closure	15	0.733	0.045	0.067	0.000	4.278
unseen_dimensions	Force closure	15	0.667	0.063	0.200	0.000	4.000
unseen_dimensions	Force closure	15	0.733	0.064	0.067	0.000	4.201
unseen_dimensions	Force closure	15	0.467	0.063	0.267	0.000	3.524
unseen_dimensions	Force closure	15	0.667	0.071	0.200	0.000	3.688
unseen_dimensions	Force closure	15	0.667	0.067	0.133	0.000	4.249
unseen_dimensions	Force closure	15	0.467	0.072	0.400	0.000	3.081
unseen_dimensions	Collision-aware FC	15	0.733	0.062	0.200	0.000	3.984
unseen_dimensions	Collision-aware FC	15	0.867	0.042	0.000	0.000	4.531
unseen_dimensions	Collision-aware FC	15	0.800	0.063	0.067	0.000	4.306
unseen_dimensions	Collision-aware FC	15	0.733	0.064	0.067	0.000	4.397
unseen_dimensions	Collision-aware FC	15	0.667	0.064	0.133	0.000	4.154
unseen_dimensions	Collision-aware FC	15	0.800	0.063	0.067	0.000	3.708

Split	Method	Episodes	Success	Slip	Drop	Collision	Energy
unseen_dimensions	Collision-aware FC	15	0.600	0.083	0.200	0.000	4.037
unseen_dimensions	Collision-aware FC	15	0.533	0.066	0.333	0.000	3.107
unseen_dimensions	CEM analytic	15	0.600	0.067	0.333	0.000	3.270
unseen_dimensions	CEM analytic	15	0.733	0.049	0.067	0.000	4.214
unseen_dimensions	CEM analytic	15	0.733	0.059	0.133	0.000	4.201
unseen_dimensions	CEM analytic	15	0.800	0.061	0.000	0.000	4.597
unseen_dimensions	CEM analytic	15	0.533	0.060	0.333	0.000	3.369
unseen_dimensions	CEM analytic	15	0.667	0.069	0.200	0.000	3.650
unseen_dimensions	CEM analytic	15	0.600	0.078	0.200	0.000	4.049
unseen_dimensions	CEM analytic	15	0.467	0.070	0.400	0.000	2.968
unseen_dimensions	Robust perturb.	15	0.733	0.064	0.267	0.000	3.562
unseen_dimensions	Robust perturb.	15	0.800	0.050	0.067	0.000	4.051
unseen_dimensions	Robust perturb.	15	0.800	0.063	0.133	0.000	4.150
unseen_dimensions	Robust perturb.	15	0.733	0.066	0.267	0.000	3.779
unseen_dimensions	Robust perturb.	15	0.667	0.064	0.200	0.000	3.846
unseen_dimensions	Robust perturb.	15	0.800	0.064	0.067	0.000	3.714
unseen_dimensions	Robust perturb.	15	0.600	0.077	0.333	0.000	3.572
unseen_dimensions	Robust perturb.	15	0.467	0.070	0.533	0.000	2.632
unseen_dimensions	MLP energy	15	0.933	0.035	0.000	0.000	2.542
unseen_dimensions	MLP energy	15	1.000	0.026	0.000	0.000	1.986
unseen_dimensions	MLP energy	15	1.000	0.042	0.000	0.000	2.078
unseen_dimensions	MLP energy	15	1.000	0.034	0.000	0.000	2.263
unseen_dimensions	MLP energy	15	0.867	0.052	0.067	0.000	2.367
unseen_dimensions	MLP energy	15	1.000	0.047	0.000	0.000	2.182
unseen_dimensions	MLP energy	15	0.933	0.035	0.000	0.000	2.179
unseen_dimensions	MLP energy	15	1.000	0.034	0.000	0.000	1.885
unseen_dimensions	HGB energy	15	1.000	0.042	0.000	0.000	2.147
unseen_dimensions	HGB energy	15	1.000	0.034	0.000	0.000	2.110
unseen_dimensions	HGB energy	15	1.000	0.046	0.000	0.000	2.050
unseen_dimensions	HGB energy	15	1.000	0.044	0.000	0.000	1.974
unseen_dimensions	HGB energy	15	1.000	0.045	0.000	0.000	2.029
unseen_dimensions	HGB energy	15	1.000	0.044	0.000	0.000	2.156
unseen_dimensions	HGB energy	15	1.000	0.037	0.000	0.000	2.105
unseen_dimensions	HGB energy	15	1.000	0.051	0.000	0.000	2.032
unseen_dimensions	Transformer	15	0.933	0.038	0.000	0.000	2.036
unseen_dimensions	Transformer	15	1.000	0.031	0.000	0.000	2.051
unseen_dimensions	Transformer	15	0.933	0.056	0.000	0.000	2.033
unseen_dimensions	Transformer	15	1.000	0.035	0.000	0.000	2.120
unseen_dimensions	Transformer	15	0.933	0.040	0.000	0.000	2.116
unseen_dimensions	Transformer	15	1.000	0.033	0.000	0.000	2.036
unseen_dimensions	Transformer	15	1.000	0.027	0.000	0.000	2.103
unseen_dimensions	Transformer	15	1.000	0.049	0.000	0.000	1.999
unseen_dimensions	Ensemble	15	1.000	0.053	0.000	0.000	1.937
unseen_dimensions	Ensemble	15	1.000	0.033	0.000	0.000	1.925
unseen_dimensions	Ensemble	15	1.000	0.049	0.000	0.000	1.961
unseen_dimensions	Ensemble	15	1.000	0.050	0.000	0.000	1.862
unseen_dimensions	Ensemble	15	1.000	0.038	0.000	0.000	2.046
unseen_dimensions	Ensemble	15	1.000	0.045	0.000	0.000	2.028
unseen_dimensions	Ensemble	15	1.000	0.055	0.000	0.000	2.057
unseen_dimensions	Ensemble	15	1.000	0.045	0.000	0.000	1.736
unseen_dimensions	Calibrated stack	15	1.000	0.034	0.000	0.000	2.299
unseen_dimensions	Calibrated stack	15	1.000	0.035	0.000	0.000	2.252
unseen_dimensions	Calibrated stack	15	1.000	0.048	0.000	0.000	2.262
unseen_dimensions	Calibrated stack	15	1.000	0.033	0.000	0.000	1.998
unseen_dimensions	Calibrated stack	15	1.000	0.039	0.000	0.000	2.203
unseen_dimensions	Calibrated stack	15	1.000	0.046	0.000	0.000	2.173
unseen_dimensions	Calibrated stack	15	1.000	0.044	0.000	0.000	2.127
unseen_dimensions	Calibrated stack	15	1.000	0.048	0.000	0.000	1.927
unseen_dimensions	Old EBM	15	0.933	0.047	0.067	0.000	2.200
unseen_dimensions	Old EBM	15	0.933	0.045	0.000	0.000	2.559
unseen_dimensions	Old EBM	15	0.933	0.050	0.000	0.000	3.085
unseen_dimensions	Old EBM	15	0.933	0.052	0.000	0.000	2.496
unseen_dimensions	Old EBM	15	0.667	0.053	0.000	0.000	2.408
unseen_dimensions	Old EBM	15	0.933	0.049	0.067	0.000	2.116
unseen_dimensions	Old EBM	15	1.000	0.035	0.000	0.000	2.414
unseen_dimensions	Old EBM	15	0.867	0.059	0.067	0.000	2.085
unseen_dimensions	EBM v5	15	1.000	0.034	0.000	0.000	2.601
unseen_dimensions	EBM v5	15	1.000	0.033	0.000	0.000	3.323
unseen_dimensions	EBM v5	15	1.000	0.037	0.000	0.000	4.036
unseen_dimensions	EBM v5	15	0.933	0.032	0.067	0.000	2.900
unseen_dimensions	EBM v5	15	0.867	0.042	0.000	0.000	3.133
unseen_dimensions	EBM v5	15	1.000	0.043	0.000	0.000	3.091
unseen_dimensions	EBM v5	15	1.000	0.041	0.000	0.000	3.038
unseen_dimensions	EBM v5	15	0.933	0.051	0.067	0.000	2.626
unseen_dimensions	Oracle grid	15	1.000	0.026	0.000	0.000	3.516
unseen_dimensions	Oracle grid	15	1.000	0.016	0.000	0.000	3.059
unseen_dimensions	Oracle grid	15	1.000	0.024	0.000	0.000	2.638
unseen_dimensions	Oracle grid	15	0.933	0.025	0.000	0.000	2.771
unseen_dimensions	Oracle grid	15	1.000	0.030	0.000	0.000	3.959

Split	Method	Episodes	Success	Slip	Drop	Collision	Energy
unseen_dimensions	Oracle grid	15	1.000	0.026	0.000	0.000	2.707
unseen_dimensions	Oracle grid	15	1.000	0.028	0.000	0.000	2.844
unseen_dimensions	Oracle grid	15	0.933	0.026	0.067	0.000	3.332
unseen_shape_family	Random	15	0.200	0.069	0.267	0.000	1.858
unseen_shape_family	Random	15	0.467	0.069	0.000	0.000	2.519
unseen_shape_family	Random	15	0.467	0.071	0.000	0.000	2.785
unseen_shape_family	Random	15	0.333	0.078	0.067	0.000	2.768
unseen_shape_family	Random	15	0.600	0.058	0.000	0.000	2.809
unseen_shape_family	Random	15	0.267	0.057	0.067	0.000	2.720
unseen_shape_family	Random	15	0.400	0.065	0.067	0.000	2.307
unseen_shape_family	Random	15	0.400	0.070	0.067	0.000	2.509
unseen_shape_family	Antipodal	15	0.200	0.058	0.000	0.000	2.924
unseen_shape_family	Antipodal	15	0.400	0.067	0.000	0.000	2.906
unseen_shape_family	Antipodal	15	0.533	0.048	0.000	0.000	3.111
unseen_shape_family	Antipodal	15	0.200	0.046	0.000	0.000	2.925
unseen_shape_family	Antipodal	15	0.533	0.071	0.000	0.000	3.375
unseen_shape_family	Antipodal	15	0.200	0.068	0.000	0.000	3.076
unseen_shape_family	Antipodal	15	0.400	0.075	0.000	0.000	2.784
unseen_shape_family	Antipodal	15	0.467	0.039	0.000	0.000	2.864
unseen_shape_family	Force closure	15	0.933	0.057	0.000	0.000	3.679
unseen_shape_family	Force closure	15	0.867	0.060	0.000	0.000	3.506
unseen_shape_family	Force closure	15	0.933	0.057	0.000	0.000	3.418
unseen_shape_family	Force closure	15	0.867	0.058	0.000	0.000	3.452
unseen_shape_family	Force closure	15	1.000	0.048	0.000	0.000	3.533
unseen_shape_family	Force closure	15	0.733	0.073	0.000	0.000	3.568
unseen_shape_family	Force closure	15	0.933	0.064	0.000	0.000	3.462
unseen_shape_family	Force closure	15	0.867	0.048	0.000	0.000	3.178
unseen_shape_family	Collision-aware FC	15	0.933	0.063	0.000	0.000	3.714
unseen_shape_family	Collision-aware FC	15	0.867	0.063	0.000	0.000	3.547
unseen_shape_family	Collision-aware FC	15	0.933	0.060	0.000	0.000	3.478
unseen_shape_family	Collision-aware FC	15	0.933	0.054	0.000	0.000	3.614
unseen_shape_family	Collision-aware FC	15	1.000	0.052	0.000	0.000	3.718
unseen_shape_family	Collision-aware FC	15	0.733	0.072	0.000	0.000	3.786
unseen_shape_family	Collision-aware FC	15	0.933	0.066	0.000	0.000	3.609
unseen_shape_family	Collision-aware FC	15	0.933	0.052	0.000	0.000	3.467
unseen_shape_family	CEM analytic	15	1.000	0.058	0.000	0.000	3.718
unseen_shape_family	CEM analytic	15	0.933	0.058	0.000	0.000	3.524
unseen_shape_family	CEM analytic	15	0.933	0.060	0.000	0.000	3.567
unseen_shape_family	CEM analytic	15	0.867	0.061	0.000	0.000	3.534
unseen_shape_family	CEM analytic	15	1.000	0.050	0.000	0.000	3.754
unseen_shape_family	CEM analytic	15	0.733	0.072	0.000	0.000	3.789
unseen_shape_family	CEM analytic	15	1.000	0.065	0.000	0.000	3.494
unseen_shape_family	CEM analytic	15	0.933	0.046	0.000	0.000	3.507
unseen_shape_family	Robust perturb.	15	1.000	0.066	0.000	0.000	3.885
unseen_shape_family	Robust perturb.	15	0.800	0.069	0.000	0.000	3.638
unseen_shape_family	Robust perturb.	15	0.933	0.056	0.000	0.000	3.604
unseen_shape_family	Robust perturb.	15	0.933	0.055	0.000	0.000	3.689
unseen_shape_family	Robust perturb.	15	1.000	0.053	0.000	0.000	3.834
unseen_shape_family	Robust perturb.	15	0.800	0.065	0.000	0.000	3.840
unseen_shape_family	Robust perturb.	15	1.000	0.064	0.000	0.000	3.675
unseen_shape_family	Robust perturb.	15	0.933	0.052	0.000	0.000	3.539
unseen_shape_family	MLP energy	15	0.933	0.048	0.000	0.000	3.342
unseen_shape_family	MLP energy	15	0.867	0.034	0.000	0.000	3.243
unseen_shape_family	MLP energy	15	0.933	0.028	0.000	0.000	2.858
unseen_shape_family	MLP energy	15	0.933	0.027	0.000	0.000	3.325
unseen_shape_family	MLP energy	15	0.800	0.049	0.000	0.000	3.458
unseen_shape_family	MLP energy	15	0.933	0.037	0.000	0.000	3.120
unseen_shape_family	MLP energy	15	0.933	0.046	0.000	0.000	3.343
unseen_shape_family	MLP energy	15	0.933	0.030	0.000	0.000	3.254
unseen_shape_family	HGB energy	15	1.000	0.048	0.000	0.000	3.497
unseen_shape_family	HGB energy	15	1.000	0.049	0.000	0.000	3.088
unseen_shape_family	HGB energy	15	1.000	0.046	0.000	0.000	3.065
unseen_shape_family	HGB energy	15	1.000	0.049	0.000	0.000	3.292
unseen_shape_family	HGB energy	15	1.000	0.042	0.000	0.000	3.399
unseen_shape_family	HGB energy	15	0.933	0.057	0.000	0.000	3.366
unseen_shape_family	HGB energy	15	1.000	0.057	0.000	0.000	3.331
unseen_shape_family	HGB energy	15	1.000	0.043	0.000	0.000	3.177
unseen_shape_family	Transformer	15	0.800	0.064	0.000	0.000	3.605
unseen_shape_family	Transformer	15	0.867	0.054	0.000	0.000	3.512
unseen_shape_family	Transformer	15	1.000	0.044	0.000	0.000	3.444
unseen_shape_family	Transformer	15	0.933	0.055	0.000	0.000	3.566
unseen_shape_family	Transformer	15	0.933	0.061	0.000	0.000	3.645
unseen_shape_family	Transformer	15	0.867	0.062	0.000	0.000	3.648
unseen_shape_family	Transformer	15	0.933	0.062	0.000	0.000	3.347
unseen_shape_family	Transformer	15	1.000	0.027	0.000	0.000	3.386
unseen_shape_family	Ensemble	15	1.000	0.062	0.000	0.000	3.447
unseen_shape_family	Ensemble	15	1.000	0.060	0.000	0.000	3.340
unseen_shape_family	Ensemble	15	1.000	0.048	0.000	0.000	3.240
unseen_shape_family	Ensemble	15	1.000	0.051	0.000	0.000	3.288

Split	Method	Episodes	Success	Slip	Drop	Collision	Energy
unseen_shape_family	Ensemble	15	1.000	0.042	0.000	0.000	3.622
unseen_shape_family	Ensemble	15	0.867	0.068	0.000	0.000	3.593
unseen_shape_family	Ensemble	15	1.000	0.055	0.000	0.000	3.150
unseen_shape_family	Ensemble	15	1.000	0.053	0.000	0.000	2.967
unseen_shape_family	Calibrated stack	15	1.000	0.050	0.000	0.000	3.586
unseen_shape_family	Calibrated stack	15	1.000	0.050	0.000	0.000	3.496
unseen_shape_family	Calibrated stack	15	1.000	0.052	0.000	0.000	3.466
unseen_shape_family	Calibrated stack	15	1.000	0.041	0.000	0.000	3.505
unseen_shape_family	Calibrated stack	15	1.000	0.047	0.000	0.000	3.716
unseen_shape_family	Calibrated stack	15	0.867	0.062	0.000	0.000	3.675
unseen_shape_family	Calibrated stack	15	1.000	0.049	0.000	0.000	3.520
unseen_shape_family	Calibrated stack	15	1.000	0.032	0.000	0.000	3.441
unseen_shape_family	Old EBM	15	0.867	0.062	0.000	0.000	3.684
unseen_shape_family	Old EBM	15	0.867	0.057	0.000	0.000	3.611
unseen_shape_family	Old EBM	15	1.000	0.052	0.000	0.000	3.480
unseen_shape_family	Old EBM	15	0.933	0.055	0.000	0.000	3.564
unseen_shape_family	Old EBM	15	0.933	0.061	0.000	0.000	3.722
unseen_shape_family	Old EBM	15	0.667	0.074	0.000	0.000	3.784
unseen_shape_family	Old EBM	15	0.933	0.066	0.000	0.000	3.430
unseen_shape_family	Old EBM	15	1.000	0.029	0.000	0.000	3.568
unseen_shape_family	EBM v5	15	1.000	0.058	0.000	0.000	3.709
unseen_shape_family	EBM v5	15	0.933	0.052	0.000	0.000	3.572
unseen_shape_family	EBM v5	15	1.000	0.045	0.000	0.000	3.598
unseen_shape_family	EBM v5	15	0.933	0.050	0.000	0.000	3.645
unseen_shape_family	EBM v5	15	1.000	0.056	0.000	0.000	3.715
unseen_shape_family	EBM v5	15	0.867	0.060	0.000	0.000	3.822
unseen_shape_family	EBM v5	15	1.000	0.053	0.000	0.000	3.559
unseen_shape_family	EBM v5	15	1.000	0.035	0.000	0.000	3.459
unseen_shape_family	Oracle grid	15	1.000	0.020	0.000	0.000	2.580
unseen_shape_family	Oracle grid	15	1.000	0.020	0.000	0.000	2.817
unseen_shape_family	Oracle grid	15	1.000	0.015	0.000	0.000	2.762
unseen_shape_family	Oracle grid	15	1.000	0.016	0.000	0.000	3.116
unseen_shape_family	Oracle grid	15	1.000	0.013	0.000	0.000	2.862
unseen_shape_family	Oracle grid	15	1.000	0.016	0.000	0.000	2.795
unseen_shape_family	Oracle grid	15	1.000	0.017	0.000	0.000	2.885
unseen_shape_family	Oracle grid	15	1.000	0.013	0.000	0.000	3.091

G ALGORITHM SKETCH

1. Generate a shared candidate set for every split, seed, and episode.
2. Score each candidate with analytic object, contact, task, collision, feasibility, and robust perturbation energies.
3. Score the same candidate with MLP, HGB, Transformer, ensemble, and scalar-energy learned models.
4. Form the calibrated learned context score.
5. Subtract weighted analytic energies and hard collision/robust penalties.
6. If the chosen candidate violates the robust-risk threshold, activate the robust analytic fallback.
7. Execute the selected candidate in MuJoCo and record success, slip, drop, collision, unsafe force, energy, and task satisfaction.
8. Repeat for all methods on the same scenario and candidate pool.

H PROOF DETAILS

Proposition 1. For each failure factor F_k , assume calibrated upper-bound cost $c_k + \epsilon_k$. Then

$$\Pr[\cup_k F_k \mid s, a] \leq \sum_k \Pr[F_k \mid s, a] \leq \sum_k c_k(s, a) + \sum_k \epsilon_k.$$

The guarantee is only as good as the calibration assumption. The frozen ablation and stress tests are empirical attacks on that assumption.

Proposition 2. If $Y(s, a) = 0$ for all $a \in A(s)$, then every deterministic or stochastic ranker over $A(s)$ has zero success probability conditional on s . Thus oracle weakness indicates candidate-set or action-parameterization failure rather than ranking failure alone.

Table 5: Pre-specified negative cases beyond the local benchmark scope.

Case	Observed failure mode	Submission implication
deformable_object	rigid-body MuJoCo energies cannot represent deformation or post-contact shape change	needs deformable benchmark before claiming general compositional grasping
occluded_geometry	candidate features assume known object extent; occlusion can make all rankers overconfident	requires perception uncertainty and real visual input
adversarial_low_friction_coating	without tactile probing, learned rankers confuse shape-compatible grasps with stable grasps	needs tactile or material sensing for deployment claims
semantic_task_ambiguity	all physical energies can be satisfied while the semantic instruction is wrong	scope is physical grasp selection, not language grounding

Proposition 3. Let M be full v5 and M_{-j} be the method with component j removed. If M_{-j} matches or exceeds M on the hostile ablation distribution within the pre-registered tolerance, the finite evidence does not identify j as necessary. The correct action is not to hide that result; it is to report it.

I WHAT WOULD MAKE THIS REVIVABLE

A credible revival needs at least four changes. First, the candidate generator must be learned or optimized jointly with the ranker; otherwise all rankers inherit the same action-support bottleneck. Second, the energies must be calibrated against held-out failure factors, not merely hand-weighted. Third, the fallback must be reported as a separate operating regime with its own success/safety frontier. Fourth, the benchmark must move to real RGB-D/tactile data or recognized public grasp benchmarks. Without those changes, compositional energy language risks becoming an attractive label for a robust analytic heuristic.

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